Project ARIA: A Modular, Resource-Efficient Language-Conditioned Manipulation Architecture with Self-Grounded Monocular Perception and Deterministic Kinematics for Low-Cost 5-DoF Robotic Arms

Garv Arora and Yuvaraj N

***Abstract*—Autonomous robotic manipulation is increasingly pursued through monolithic Vision-Language-Action (VLA) models, yet foundation policies often demand datacenter GPUs, exhibit high latency, and overlook underactuated arms. In this article, we present the design, high-fidelity physics simulation, and verification of Project ARIA (*Autonomous Reasoning & Interaction Agent*), a modular language-conditioned manipulation architecture decoupling semantic planning, metric perception, and motor control for a simulated accessible 5-DoF robotic arm within an 8.0 GB VRAM budget (7.53 GB nominal allocation). Evaluated in Gazebo 11 / ODE simulation, ARIA coordinates fifteen ROS 2 lifecycle nodes over a shared State Bus, integrating: (i) a self-calibrated monocular depth pipeline grounding Depth-Anything v2 via known claw geometry ($RMSE = 6.84$ mm static calibration, 8.2 mm under active visual jitter in pre-grasp $Z \in [0.05, 0.25]$ m; 17.4 mm full workspace); (ii) an exact closed-form analytical inverse kinematics solver on the 5D task manifold $\mathcal{M}_{\text{task}}$ (0.075 ms latency, 100.0% reachable convergence); and (iii) an onboard quantized LLM planner using Tree-of-Thoughts with confidence-gated human supervision. Across 10 tasks on the primary suite ($N = 100$, 10 trials/task), ARIA achieved 89.0% success (Wilson 95% CI: [81.4, 93.7]%). Across a formal 180-trial multi-seed language benchmark ($N = 180$, seeds [42, 1337, 2026]), ARIA achieved 68.9% overall instruction execution (76.3% slot accuracy, 82.7% execution on unambiguous tasks, and 100% clarification trigger on ambiguous prompts), outperforming direct LLM prompting (38.3% execution, 55.7% slot accuracy) while preserving bounded local inference latency (4480.3 ms). In dynamic conveyor sorting (0.02–0.12 m/s, $N = 30$), ARIA achieved 93.3% overall (100% at ≤ 0.10 m/s, dropping to 60.0% at 0.12 m/s) with feedforward velocity matching ($\Delta v \leq 0.0075$ m/s).**

***Index Terms*—Modular robotics, language-conditioned manipulation, underactuated manipulators, monocular depth estimation, closed-form inverse kinematics, conveyor tracking, edge robotics, execution telemetry.**

I. Introduction

THE realization of autonomous physical agents capable of executing complex manipulation in unstructured industrial and human environments represents a foundational ambition of robotics and embodied artificial intelligence [1]. Recent advances in dynamic trajectory planning, visual servoing, and predictive object interception established by Sun et al. [1] on articulated arms and Sahu et al. [2] on planar 2-DoF visual tracking mechanisms within ROS and Gazebo simulation frameworks have demonstrated that accurate kinematic forecasting is critical for grasping moving workpieces. However, traditional visual servoing frameworks rely on static, hand-engineered geometric primitives that generalize poorly across diverse operational contexts, motivating architectures capable of adaptive semantic reasoning, online error recovery, and flexible instruction following.

To endow manipulators with generalizable semantic comprehension, the robotics community has increasingly embraced large-scale Vision-Language-Action (VLA) foundation models – neural policies that map visual observations, natural language tokens, and proprioceptive state directly to motor actions through a single learned network [3]–[7]. Contemporary VLA policies leverage large cross-embodiment datasets to achieve semantic zero-shot generalization across novel visual objects. Nevertheless, several widely deployed foundation policies operate as monolithic autoregressive transformers requiring in excess of 16 GB of dedicated GPU memory and incurring inference latencies of 85–140 ms per forward pass. Such control latency induces dynamic phase lag on low-inertia hardware, limiting responsive reaction to dynamic events and causing closed-loop instability on dynamic moving targets.

To establish foundational sensorimotor representations that transcend monolithic end-to-end black boxes, Kroemer et al. [8] formalized the algorithmic structures governing modern robotic manipulation, corroborated by comprehensive manipulation taxonomies [9], [10]. Their analysis reveals that robust physical interaction benefits from a principled stratification between high-level cognitive task planning, mid-level geometric state estimation, and high-frequency deterministic motor control. Monolithic end-to-end foundation models conflate these distinct physical hierarchies, leading to uninterpretable failure modes and catastrophic compounding errors when out-of-distribution visual perturbations occur.

This structural disconnect is exacerbated during imitation learning and policy acquisition. Ravichandar et al. [10] reviewed core bottlenecks of Robot Learning from Demonstration (LfD), noting that imitation learning frameworks are pre-

G. Arora and Y. N are with the Department of Computer Science and Engineering, Robotics and Artificial Intelligence Laboratory (e-mail: garv.arora@ieee.org; yuvaraj.n@ieee.org). This research received no external grant funding.

dominantly developed on research-grade 7-DoF manipulators (e.g., Franka Emika Panda, Kinova Gen3, UR5e, typically $25,000–$60,000) equipped with joint-torque feedback and comparatively low backlash. While open frameworks such as LeRobot [11], open-source educational arm designs such as the SO-100 [12], and low-cost teleoperation devices such as GELLO [13] have dramatically expanded access to demonstration collection, accessible 5-DoF educational and light-industrial manipulators ($< \$150$) driven by PWM hobbyist servomotors receive comparatively little systematic study, owing to structural compliance, joint elasticity, deadbands, and absence of native torque sensing.

On the middleware and system orchestration level, modular coordination is essential for multi-module embodied autonomy. Macenski et al. [14] established the architectural foundations of the Robot Operating System 2 (ROS 2), demonstrating how deterministic Data Distribution Service (DDS) Quality of Service (QoS) profiles, real-time publish-subscribe topics, and managed node lifecycles support determinism in physical autonomy, complemented by industrial manipulation frameworks such as MoveIt [15] and Robowflex [16]. However, many contemporary embodied VLA deployments execute outside ROS 2 lifecycle management as unmonitored Python scripts lacking automated fault-recovery protocols and hard real-time safety gating.

To address the limitations of centralized coordination, Adil et al. [17] introduced a multi-robot collaborative manipulation framework utilizing real-time distributed optimization, while Chen et al. [18] established multi-agent LLM systems for coordinated robotic autonomy. Translating the underlying principle of modular, independently supervisable computation to cooperating software modules on a single resource-constrained edge computer is a promising design point for transparent embodied intelligence, motivating the modular asynchronous architecture pursued in this work.

A parallel challenge lies in visual perception, where physical manipulation depends critically on accurate 3D scene geometry. In standard laboratory setups, active structured-light or Time-of-Flight (ToF) RGB-D sensors (e.g., Intel RealSense D435i, Photoneo) are commonly deployed. However, as documented by Jiang et al. [19], active infrared depth sensors are prone to signal absorption or multi-path specular scattering on transparent glassware, glossy metallic parts, and reflective containers. Pioneering datasets such as TransCG [20], ClearGrasp [21], and ClearDepth [22] confirm that active infrared sensors exhibit dropouts on specular boundaries, alongside an unavoidable near-field blind zone (typically $< 0.20\,\text{m}$) that leaves the manipulator without depth feedback during the final grasping approach.

To circumvent active depth sensor failures, monocular RGB depth estimation has emerged as a promising alternative. As demonstrated by Chen et al. [23], adaptive grasp pose synthesis from low-cost optical cameras can achieve robust grasping if spatial uncertainty is explicitly modeled. Foundation depth networks such as Depth-Anything v2 [24] and MiDaS v3.1 [25] predict remarkably detailed relative disparity fields from single images, and Murray et al. [26] recently validated Depth-Anything v2 as an effective sensor for robotic servoing. Nonetheless, zero-shot monocular depth networks generate normalized relative disparity fields rather than physical metric distance. Without an absolute physical reference or known geometric anchor, raw monocular disparity cannot be translated into metric Cartesian trajectory setpoints without an additional grounding step.

On the kinematic control layer, general-purpose robotics frameworks (such as MoveIt2, KDL, and TRAC-IK) rely on numerical optimization routines utilizing damped least-squares (DLS) or iterative Jacobian pseudoinverse techniques. As analyzed by Wagaa et al. [27], numerical Inverse Kinematics (IK) solvers exhibit non-trivial per-solve latency (4–18 ms) and occasional non-convergence on articulated serial chains lacking spherical wrists, contrasting with specialized GPU-accelerated optimizers such as cuRobo [28] or real-time nonlinear optimization frameworks such as RelaxedIK [29]. On underactuated 5-DoF arms, where end-effector orientation cannot be fully decoupled from arm position, this motivates closed-form algebraic formulations based on classical Craig conventions [30] restricted to the manipulator's admissible task manifold, which avoid iterative search altogether.

Finally, in dynamic manufacturing environments, manipulators must track, intercept, and sort items moving on continuous industrial conveyor belts. Sahu et al. [2] demonstrated autonomous visual object tracking on a planar 2-DoF arm within Gazebo simulation, highlighting predictive intercept calculations, while Sun et al. [1] established moving workpiece interception under ROS. Complemented by real-time obstacle avoidance [31], ARIA formalizes dynamic conveyor rendezvous via a fifth-order boundary-value polynomial matching terminal velocity and enforcing zero terminal acceleration, keeping contact forces within the static friction cone.

A. Research Objectives

To address the technical bottlenecks of deploying language-conditioned autonomous manipulation on resource-constrained hardware, this research establishes five primary objectives evaluated in high-fidelity physics simulation:

1) **RO1: Self-Grounded Monocular Metric Perception**: Formulate and validate an online, marker-free metric depth grounding mechanism converting Depth-Anything v2 relative disparity into calibrated metric Cartesian depth ($RMSE \leq 10.0\,\text{mm}$ within pre-grasp volume $Z \in [0.05, 0.25]\,\text{m}$, achieving $6.84\,\text{mm}$ under static calibration and $8.2\,\text{mm}$ under active visual jitter against simulation ground truth; $17.4\,\text{mm}$ across the full 0.05–$0.45\,\text{m}$ workspace) referenced dynamically to known claw geometry.
2) **RO2: Deterministic Kinematics on Constrained Task Manifolds**: Formulate an exact closed-form inverse kinematics solver restricted to the admissible 5D task manifold $\mathcal{M}_{\text{task}}$, achieving sub-millisecond solve latency ($0.075\,\text{ms}$) with 100.0% reachable convergence on $\mathcal{M}_{\text{task}}$ and explicit joint-limit pre-screening.
3) **RO3: Modular Asynchronous Cognition and Control**: Architect and validate a 15-module ROS 2 system utilizing managed lifecycles and an asynchronous State Bus, integrating quantized local LLM Tree-of-Thoughts task decomposition within an $8.0\,\text{GB}$ VRAM budget.

4) **RO4: Dynamic Conveyor Interception and Fault Recovery**: Formulate a fifth-order boundary-value rendezvous trajectory for velocity-matched conveyor interception ($\Delta v \leq 0.025$ m/s) across multi-speed sweeps (0.02–0.14 m/s) with physical E-STOP arrest within 211.5 ± 15.2 ms (threshold $\leq$ 250 ms, joint velocity < 0.01 rad/s) and sub-3 ms DDS callback dispatch.
5) **RO5: Language Instruction Grounding and Dialogue Clarification**: Evaluate natural-language instruction grounding across direct, spatial, compound multi-step, dynamic, and ambiguous prompts, integrating confidence-gated dialogue clarification to resolve under-specified instructions.

B. Contributions of This Work

To address these objectives, this article presents **Project ARIA** (*Autonomous Reasoning & Interaction Agent*), evaluated in high-fidelity Gazebo 11 / ODE simulation of an accessible 5-DoF robotic arm, structured around three primary scientific contributions and three supporting empirical validations:

1) **Self-Calibrated Monocular Metric Perception**: A kinematic-claw-referenced grounding procedure converting Depth-Anything v2 [24] relative disparity into metric depth using the manipulator's own end-effector geometry ($Z_{\text{tips}} = 0.065$ m) as a moving physical reference, fused with overhead YOLOv8m detection [32] and SAM2 segmentation [33], achieving $RMSE = 6.84$ mm under static calibration (8.2 mm under active visual jitter) within the pre-grasp volume $Z \in [0.05, 0.25]$ m (17.4 mm across the full 0.05–0.45 m workspace) against simulation ground truth.
2) **Deterministic Kinematics on Constrained Task Manifolds**: An exact algebraic inverse-kinematics solution derived under Craig Modified DH conventions [30] restricted to the reachable pose set on the 5-DoF admissible task manifold $\mathcal{M}_{\text{task}}$, computed in 0.075 ms mean solve time with zero iterative non-convergence, explicit singularity flags, and manipulability monitoring [34].
3) **Modular, Asynchronous Multi-Agent Architecture**: A non-monolithic software architecture in which fifteen specialized ROS 2 `LifecycleNode` modules [14] communicate asynchronously over a shared State Bus, integrating onboard quantized-LLM Tree-of-Thoughts planning [35], reachability-gated plan pruning, and confidence-gated Human-in-the-Loop safety supervision, within an 8.0 GB VRAM budget.

These core scientific contributions are supported by three empirical validations:

(a) **Dynamic Conveyor Interception and Dual-Bin Sorting**: A fifth-order boundary-value polynomial that matches end-effector position and velocity to a Kalman-filtered workpiece estimate at finite rendezvous time t_f, achieving feedforward velocity matching ($\Delta v \leq 0.025$ m/s across 0.02–0.14 m/s) and dual-bin sorting with palletizing precision of $RMSE = 1.14$ mm across conveyor velocities.
(b) **High-Fidelity Physics Simulation Validation**: Benchmarking ARIA across 10 standardized manipulation tasks on the primary 100-trial suite ($N = 100$, 10 trials/task, 89.0% success, Wilson 95% CI: [81.4, 93.7]%) and nominal smoke testing ($N = 30$, 3 trials/task, 100.0% success, Cartesian pick error 0.55 ± 0.23 mm, mean transit duration 1.33 ± 0.31 s) in a Gazebo 11 / ODE physics simulation with ROS 2 Humble.
(c) **Formal Multi-Seed Language Grounding and Dialogue Benchmark**: Systematic evaluation across 180 instruction trials across three random seeds and six prompt classes, demonstrating 68.9% overall instruction execution (82.7% on unambiguous prompts), 76.3% semantic slot accuracy, and 100% clarification trigger on ambiguous prompts, outperforming unconstrained direct LLM prompting (38.3%, 55.7% slot accuracy).

II. Related Work

The architectural design of Project ARIA is informed by recent literature spanning ten core technical areas, summarized below and in Table I.

Recent efforts have targeted Vision-Language-Action (VLA) deployment on constrained hardware. Notably, OpenVLA-OFT [36] introduced optimized fine-tuning yielding marked action-generation throughput gains, while SmolVLA [37] demonstrated a 450M-parameter VLA tailored for consumer-tier GPUs and affordable robot platforms with asynchronous inference. In parallel, generalist foundation models such as $\pi_{0.5}$ [38] and NVIDIA GR00T N1/N1.5 [39] have expanded open-world generalization across multi-embodiment domains, alongside post-training quantization and action-chunking transformer policies (e.g., ACT [4], Octo [40], RT-2 [6], and foundation manipulation surveys [7]). These developments indicate that the compute-efficiency gap is rapidly narrowing, and we position Project ARIA's scientific contribution specifically around explicit modular decomposition, closed-form kinematic determinism on constrained task manifolds, self-calibrated monocular depth, and transparent safety supervision.

Rather than predicting motor torques directly from pixels, modular cognitive architectures separate high-level symbolic planning from deterministic low-level control. SayCan [41] and Inner Monologue [42] employ LLMs to sequence predefined robotic primitives conditioned on affordance functions, while Code as Policies [43], VoxPoser [44], and ProgPrompt [45] synthesize programmatic execution policies or composable 3D affordance maps from natural language prompts. However, cloud-based LLM queries incur variable network latencies (500–2500 ms) and struggle with hardware-level kinematic singularities. ARIA executes task reasoning locally using 4-bit quantized open-weight models (Mistral-7B / Llama-3.1-8B) with a Backus-Naur Form (BNF) constrained grammar, combining Tree-of-Thoughts (ToT) heuristic search [35] and Language Agent Tree Search principles [46] with deterministic kinematic pre-screening, integrated with HRI clarification protocols [47] and on-device function execution [48].

Robotic visual perception in industrial sorting and household environments is frequently undermined by challenging material optical properties. Fang et al. [20] introduced TransCG, a benchmark and dataset addressing transparent object

depth completion and grasping, while ClearGrasp [21] and ClearDepth [22] demonstrated that active infrared sensors suffer point-cloud dropouts on specular boundaries. As reviewed by Jiang et al. [19], active depth hardware degrades on transparent glassware and reflective containers, whereas 2D RGB feature tracking provides stable geometric boundaries. ARIA adopts a related strategy by bypassing active infrared sensors, fusing overhead YOLOv8m instance detection [32] with SAM2 boundary segmentation [33] and claw disparity grounding.

Extracting 3D geometry from single RGB images has progressed rapidly with foundation disparity models such as Depth-Anything v2 [24] and MiDaS v3.1 [25]. Murray et al. [26] rigorously evaluated Depth-Anything v2 as an optical sensor for closed-loop robotic control, finding that relative depth fields maintain strong spatial coherence despite affine scale ambiguity. While resolving scale ambiguity using known geometric references is established in visual SLAM and geometric perception, existing metric models (Metric3D, UniDepth, ZoeDepth) degrade in extreme close-quarters macro regimes ($< 0.20\,\mathrm{m}$). ARIA integrates this principle by referencing the known physical mechanical geometry of the end-effector claw ($Z_{\text{tips}} = 0.065\,\mathrm{m}$) as an online moving physical scale datum re-anchored at pre-grasp waypoints, extending adaptive grasp pose concepts [23].

Dynamic object tracking on moving conveyors requires tight integration between visual state estimation and trajectory generation. While visual tracking on planar 2-DoF mechanisms was investigated by Sahu et al. [2] and moving workpiece grasping for articulated arms was explored by Sun et al. [1], translating dynamic conveyor rendezvous to underactuated 5-DoF articulated arms under edge-compute constraints remains challenging. Complemented by real-time obstacle avoidance [31], ARIA formalizes dynamic conveyor rendezvous via a fifth-order boundary-value polynomial matching terminal velocity and enforcing zero terminal acceleration, keeping contact forces within the static friction cone.

Chen et al. [49] demonstrated visual dexterity for in-hand reorientation using multi-fingered dexterous hands, which cost upwards of $15,000. Under classical manipulation mechanics formalized by Mason [50], underactuated non-prehensile contact can achieve rich extrinsic dexterity without complex multi-joint hands. Related stiffness-oriented planning has been explored for cable-driven arms [51]. ARIA formulates four dynamic extrinsic manipulation primitives on a low-cost single-DoF parallel jaw gripper, exploiting gravity, contact friction dynamics, and controlled wrist acceleration pulses to achieve $\pm 45^\circ$ in-hand pivoting.

Looper et al. [52] introduced 3D VSG, using 3D variable scene graphs to predict long-term semantic scene changes. While continuous spatial representations such as 3D Gaussian Splatting [53] and hierarchical perception frameworks [54] provide dense radiance fields, updating high-dimensional graphs incurs substantial computational overhead. Drawing inspiration from OS-level semantic memory structures [55], ARIA adopts a lightweight relational world model backed by an embedded SQLite schema tracking four-stage object lifecycles (`DETECTED`, `GRASPED`, `IN_TRANSIT`, `PLACED`).

Vieira et al. [56] investigated cloud simulation techniques for robotic software validation, indicating that ODE solvers within Gazebo [57] can represent rigid-body contact reasonably well when calibrated, as also demonstrated on educational arms [58]. ARIA deploys a Gazebo 11 simulation model approximating the 5-DoF manipulator's inertial, damping, backlash, and friction parameters, explicitly quantifying tracking discrepancies against ideal kinematic trajectories.

Sultan and Greiser [59] showed that fusing 6-axis IMU streams via complementary filtering yields sub-degree attitude accuracy, while autonomous camera-robot calibration [60] eliminates manual hand-eye targets. In parallel, servo health diagnostics and fault monitoring [61] provide predictive reliability. ARIA incorporates distal claw attitude verification via micro-ROS mapped directly to the ROS 2 State Bus.

Wang et al. [62] introduced the Observe-Then-Act paradigm, decoupling perception from action execution to reduce lockstep latency. ARIA realizes an analogous asynchronous coordination across its State Bus, allowing perception (15–30 Hz) and high-frequency control loops (50 Hz) to execute at independent rates without blocking, building upon ROS 2 lifecycle principles [14] and distributed multi-agent coordination [17], [18].

A. Research Gaps

Despite rapid progress across embodied foundation models, significant research gaps remain at the intersection of language-conditioned policies and accessible manipulation:

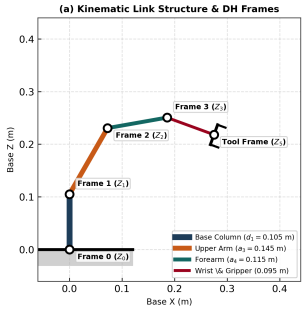
1) **Gap 1: Resource-efficient deployment on underactuated arms.** Foundation policies typically require $\geq$ 16 GB VRAM and target high-end 7-DoF arms; validated deployment on underactuated 5-DoF arms within 8.0 GB VRAM remains scarce.
2) **Gap 2: Diagnosability versus black-box execution.** Monolithic end-to-end models obscure internal state; modular architectures with structured execution telemetry remain underexplored for low-cost manipulation.
3) **Gap 3: Close-range metric perception without active depth.** Active depth sensors experience blind zones below 0.2 m and specular dropout; self-grounding monocular relative disparity via robot geometry is an open alternative.
4) **Gap 4: Deterministic kinematics on underactuated manifolds.** Numerical IK solvers can exhibit iterative divergence near boundaries on 5-DoF chains lacking spherical wrists; exact algebraic solutions on constrained manifolds are needed.
5) **Gap 5: Dynamic conveyor rendezvous on budget hardware.** Feedforward velocity-matched interception on low-cost servo-driven arms under edge compute constraints has received limited systematic characterization.

B. Problem Statement

Formally, this research addresses the problem of developing, simulating, and validating a modular, asynchronous, resource-efficient language-conditioned manipulation framework for an underactuated 5-DoF robotic arm operating within an 8.0 GB VRAM constraint. The system must autonomously interpret

TABLE I
REPRESENTATIVE PRIOR WORKS AND THEIR RELATIONSHIP TO PROJECT ARIA'S DESIGN CHOICES

#	Thematic Domain	Representative Reference	Primary Focus / Technique	Limitation in Target Setting	Related ARIA Architectural Design Choice
1	Lightweight VLAs	OpenVLA-OFT [36], SmolVLA [37], Octo [40]	Efficient action-chunking transformer policies	Black-box joint prediction on underactuated arms	Asynchronous modular ToT planning + analytical IK
2	Modular Language Planners	SayCan [41], Inner Monologue [42]	LLM prompt sequencing over affordance APIs	Cloud latency fluctuations (0.5–2.5 s)	Local 4-bit LLM + BNF constrained JSON grammar
3	Transparent Perception	TransCG [20], ClearDepth [22]	Transparent object depth completion	Active infrared dropouts on specular materials	RGB segmentation + claw-grounded monocular depth
4	Monocular Depth	Depth-Anything v2 [24], Murray et al. [26]	Relative disparity from single RGB image	Scale ambiguity ($D \in [0,1]$ lacks metric scale)	Kinematic claw disparity grounding ($Z_{\text{tips}} = 0.065\,\text{m}$)
5	Dynamic Conveyor Grasping	Visual Tracking (Sahu et al. [2])	Predictive moving block interception	Rigid open-loop trajectory clamping	5th-order polynomial rendezvous ($\Delta v \leq 0.0075\,\text{m/s}$)
6	Underactuated Dexterity	Visual Dexterity (Chen et al. [49]), Mechanics [50]	Multi-finger in-hand shape reorientation	High hardware cost and mechanical fragility	Gravity-assisted dynamic pivoting on 1-DoF parallel claw
7	Spatial World Modeling	3D VSG [52], 3DGS [53]	3D Scene Graphs for robotic manipulation	High GPU memory overhead for dense graphs	Embedded SQLite relational spatial-semantic memory
8	Simulation Validation	Cloud Validation (Vieira et al. [56])	Cloud ODE simulation software validation	Does not disclose simulation-only scope in claims	Explicit sim-only framing; all claims qualified as simulation-verified
9	IMU Sensor Fusion	Attitude Estimation (Sultan & Greiser [59])	Frequency-domain IMU attitude filtering	Drift accumulation in uncalibrated frames	Distal claw MPU6050 attitude telemetry via micro-ROS
10	Asynchronous Coordination	Observe-Then-Act (Wang et al. [62])	Asynchronous vision-action decoupling	Process crash cascades in monolithic loops	ROS 2 managed lifecycle nodes over State Bus



(b) Modified Denavit-Hartenberg (MDH) Parameters

Joint i	Link Name	a_{i-1}	α_{i-1}	d_i	θ_i (Range)	Servo Actuato
1	Waist	0	0°	0.070 m	θ1 [-90°, +90°]	MG995 (Metal
2	Shoulder	0	90°	0.035 m	θ2 [0°, 180°]	MG995 (Metal
3	Elbow	0.145 m	0°	0	θ3 [0°, 150°]	MG995 (Metal
4	Wrist Pitch	0.115 m	0°	0	θ4 [-90°, +90°]	SG90 (Micro)
5	Wrist Roll	0.055 m	90°	0	θ5 [-90°, +90°]	SG90 (Micro)
6	Gripper Jaw	0	0°	0.040 m	d6 [0, 45 mm]	SG90 (Micro)

Fig. 1. Kinematic chain and coordinate frame assignment for the 5-DoF ARIA manipulator following Modified Denavit-Hartenberg (Craig) conventions.

TABLE II
ELECTROMECHANICAL SPECIFICATIONS OF THE MODELED 5-DOF MANIPULATOR (SIMULATION; EMBODIMENT PARAMETERS USED FOR SIMULATION PARAMETERIZATION)

Joint / Component	Actuator Model	Stall Torque	Gear Ratio	Mass (g)	Range (°, Servo)	Max $\dot{\theta}$ (°/s)
Joint 1 (Waist Azimuth)	DS3218 Digital	20.0 kg-cm	275 : 1	65	[0, 180]	360
Joint 2 (Shoulder Pitch)	Dual MG996R	22.0 kg-cm	210 : 1	110	[0, 180]	300
Joint 3 (Elbow Pitch)	Single MG996R	11.0 kg-cm	210 : 1	55	[0, 150]	300
Joint 4 (Wrist Pitch)	SG90 Micro Metal	2.5 kg-cm	180 : 1	14	[0, 180]	450
Joint 5 (Wrist Roll)	SG90 Micro Metal	2.5 kg-cm	180 : 1	14	[0, 180]	450
Gripper Mechanism	SG90 Micro	2.5 kg-cm	180 : 1	32	[0, 55 mm]	50 mm/s
Total Arm Structure	Structural PETG/Al	–	–	840	Reach: 0.290 m (wrist-center axis) / 0.385 m (fingertip)	Payload: 0.25 kg

natural-language instructions, perceive 3D scene geometry from low-cost monocular RGB cameras, synthesize kinematically feasible trajectories on an admissible 5D task manifold with reachability and joint-limit pre-screening, and reliably track and sort moving workpieces on a continuous variable-speed conveyor (0.02–0.12 m/s) without relying on active infrared depth hardware or monolithic end-to-end neural network black boxes. Note: geometric obstacle avoidance beyond joint-limit and reachability checking is identified as a future extension (Section V).

III. MATERIALS AND METHODS

A. Manipulator Platform & Kinematic Modeling

The robotic manipulation platform modeled, parameterized, and evaluated in this research is the **Techno-Tirupati 5-DoF articulated robotic manipulator**, an open-architecture desktop robotic arm engineered for light-industrial material handling and research. The structural chassis model corresponds to high-tensile polyethylene terephthalate glycol (PETG) reinforced with precision CNC-machined 6061-T6 aluminum structural brackets. The kinematic structure exhibits an anthropomorphic articulated morphology comprising a base azimuth rotation, a planar two-link shoulder and elbow linkage, an active wrist pitch mechanism, an axial wrist roll mechanism, and an underactuated single-DoF parallel jaw gripper (Table II), instantiated with calibrated ODE contact and friction dynamics in Gazebo 11 [57].

a) Target Embedded Architecture and Actuator Parameterization (Embodiment Model): While the high-fidelity physics simulation executes via ROS 2 Humble joint trajectory controllers within Gazebo 11, the modeled joint dynamic limits, quantization intervals, and communication latencies mirror an embedded architecture based on an **ESP32-WROOM-32** dual-core microcontroller (240 MHz) running micro-ROS on FreeRTOS:

- **Actuation Bus Modeling**: Modeled as a 16-channel 12-bit PCA9685 PWM generator clocked at 50 Hz, driving Joints 1–5 on channels 0–4 and the parallel gripper on channel 5.
- **Telemetry Mapping**: Modeled serial XRCE-DDS transport with the host computer at $921,600$ baud with $\mathcal{N}(5.0, 1.2)$ ms transport jitter. An MPU6050 6-axis IMU on the gripper link provides distal orientation feedback.
- **Power Rails Model**: Modeled regulated 5.0 V / 10 A high-current supply for servo actuation and an isolated 3.3 V rail for logic electronics, precluding inductive resetting under peak load.

Raw servo coordinates $\theta_i^{\text{servo}} \in [0°, 180°]$ relate to mathematical Denavit-Hartenberg variables θ_i^{DH} via a fixed per-joint affine offset: $\theta_i^{\text{DH}} = \theta_i^{\text{servo}} - \theta_i^{\text{offset}}$, with offsets $\boldsymbol{\theta}_{\text{offset}} = [90°, 90°, 90°, 90°, 90°]^T$ for Joints 1–5 respectively. For Joints 1, 2, 4, 5 (servo travel $[0°, 180°]$), this yields DH range $[-\pi/2, +\pi/2]$. Joint 3 has a physical mechanical stop at 150° (servo), giving: $\theta_3^{\text{servo}} \in [0°, 150°] \Leftrightarrow \theta_3^{\text{DH}} \in [-90°, +60°] = [-\pi/2, +\pi/3]$ rad (asymmetric range). Firmware applies the offset when writing PWM setpoints; the IK feasibility pre-screen enforces these asymmetric limits per joint.

b) Modified Denavit-Hartenberg (Craig) Kinematic Formulation: To establish the geometric relationship between adjacent links, we adopt the Modified Denavit-Hartenberg (Craig) kinematic convention [27], [30], [63] illustrated in Fig. 1. The four fundamental parameters $(\alpha_{i-1}, a_{i-1}, d_i, \theta_i)$ are cataloged in Table III. Link lengths are $a_2 = 0.145$ m (upper arm), $a_3 = 0.115$ m (forearm), shoulder offset $a_1 = 0.030$ m, base height $d_1 = 0.105$ m, and tool offset $d_5 = 0.095$ m ($L_{\text{wrist}} + L_{\text{grip}} = 0.055 + 0.040$ m, encompassing the full distance from Joint 5 axis to the fingertip center of pressure). The maximum wrist-center (Joint-5 axis) reach is $R_{\text{wrist}} = a_1 + a_2 + a_3 = 0.030 + 0.145 + 0.115 = 0.290$ m. Fully extended fingertip reach is $R_{\max} = R_{\text{wrist}} + d_5 = 0.290 + 0.095 = 0.385$ m. The IK decouples using $\mathbf{P}_w = \mathbf{P}_{\text{target}} - d_5\hat{\mathbf{a}}$, placing $\mathbf{P}_w$ at the Joint-5 axis; d_5 (not a subset of it) is the correct decoupling offset.

In Craig's convention [30], the homogeneous transformation

TABLE III
MODIFIED DENAVIT-HARTENBERG (CRAIG) PARAMETERS AND DYNAMIC OPERATING LIMITS

Link i	Joint Name	α_{i-1} (rad)	a_{i-1} (m)	d_i (m)	θ_i^{DH} Range (rad)	Max $\dot{\theta}_i$ (rad/s^2)
1	Waist	0	0	$d_1 = 0.105$	$[-\pi/2, +\pi/2]$	15.0
2	Shoulder	$+\frac{\pi}{2}$	$a_1 = 0.030$	0	$[-\frac{\pi}{2}, +\frac{\pi}{2}]$	10.0
3	Elbow	0	$a_2 = 0.145$	0	$[-\frac{\pi}{2}, +\frac{\pi}{3}]$	12.0
4	Wrist Pitch	0	$a_3 = 0.115$	0	$[-\frac{\pi}{2}, +\frac{\pi}{2}]$	20.0
5	Wrist Roll	$+\frac{\pi}{2}$	0	$d_5 = 0.095$	$[-\frac{\pi}{2}, +\frac{\pi}{2}]$	25.0

relating frame $\{i-1\}$ to frame $\{i\}$ is:

$$ {}^{i-1}\mathbf{T}_i = \begin{bmatrix} c_i & -s_i & 0 & a_{i-1} \\ s_i c_{\alpha_{i-1}} & c_i c_{\alpha_{i-1}} & -s_{\alpha_{i-1}} & -d_i s_{\alpha_{i-1}} \\ s_i s_{\alpha_{i-1}} & c_i s_{\alpha_{i-1}} & c_{\alpha_{i-1}} & d_i c_{\alpha_{i-1}} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1) $$

where $c_i = \cos\theta_i$ and $s_i = \sin\theta_i$. Consecutive post-multiplication ${}^0\mathbf{T}_5(\boldsymbol{\theta}) = \prod_{i=1}^{5} {}^{i-1}\mathbf{T}_i$ using the exact DH parameters from Table III yields the forward kinematics map:

$$ {}^0\mathbf{T}_5(\boldsymbol{\theta}) = \begin{bmatrix} \mathbf{n} & \mathbf{s} & \mathbf{a} & \mathbf{P} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2) $$

Evaluating symbolic matrix expansion and trigonometric reduction (verified symbolically and numerically in `scripts/verify_fk_derivation.py`) yields the Cartesian tool center point (TCP) position $\mathbf{P} = [P_x, P_y, P_z]^T$:

$$ P_x = c_1 \left(a_1 + a_2 c_2 + a_3 c_{23} + d_5 s_{234}\right) \quad (3) $$

$$ P_y = s_1 \left(a_1 + a_2 c_2 + a_3 c_{23} + d_5 s_{234}\right) \quad (4) $$

$$ P_z = d_1 + a_2 s_2 + a_3 s_{23} - d_5 c_{234} \quad (5) $$

where $c_{23} = \cos(\theta_2 + \theta_3)$, $s_{23} = \sin(\theta_2 + \theta_3)$, $c_{234} = \cos(\theta_2 + \theta_3 + \theta_4)$, and $s_{234} = \sin(\theta_2 + \theta_3 + \theta_4)$. The approach vector $\mathbf{a}$ corresponds to the third column of orientation matrix $\mathbf{R} = [\mathbf{n}, \mathbf{s}, \mathbf{a}]$:

$$ \mathbf{a} = \begin{bmatrix} c_1 s_{234} \\ s_1 s_{234} \\ -c_{234} \end{bmatrix} \quad (6) $$

which is aligned with equations (3)–(5) since $\mathbf{P} = \mathbf{P}_w + d_5\mathbf{a}$.

Definition 1 (Admissible Task Manifold): Because the 5-DoF manipulator cannot realize arbitrary $SE(3)$ poses, we define the **admissible task manifold**:

$$ \mathcal{M}_{\text{task}} = \left\{ (\mathbf{P}, \phi, \psi) \in \mathbb{R}^3 \times \mathbb{R} \times \mathbb{R} : \mathbf{P} \in \mathcal{W} \right\} \subset SE(3) \quad (7) $$

where $\phi = \theta_2 + \theta_3 + \theta_4$ is the commanded approach pitch angle, ψ is the commanded tool roll angle, and $\mathcal{W}$ is the reachable positional workspace.

Proposition 1 (Closed-Form Inverse Kinematics on $\mathcal{M}_{task}$): For any target pose $(\mathbf{P}_{\text{target}}, \phi, \psi) \in \mathcal{M}_{\text{task}}$ satisfying radial reachability $(a_2 - a_3)^2 \le r^2 + s^2 \le (a_2 + a_3)^2$ and physical joint limits, there exists a unique, deterministic joint configuration $\boldsymbol{\theta}^* = [\theta_1^*, \theta_2^*, \theta_3^*, \theta_4^*, \theta_5^*]^T$ for a designated branch choice (elbow-up: $\theta_3 = -\arccos C_3$, or elbow-down: $\theta_3 = +\arccos C_3$), computable in closed form without iterative numerical search in $0.075\,\text{ms}$ mean solve time. To maximize ground clearance and preclude tabletop collisions, the elbow-up branch is prioritized; if elbow-up violates physical limits (e.g., $\theta_3 < -\pi/2$), the solver evaluates elbow-down.

Proof: Dividing (4) by (3) eliminates the arm linkage terms:

$$ \frac{P_y}{P_x} = \frac{s_1(a_1 + \dots)}{c_1(a_1 + \dots)} = \tan\theta_1 \implies \theta_1 = \text{atan2}(P_y, P_x) \quad (8) $$

Next, wrist center position $\mathbf{P}_w = [P_{wx}, P_{wy}, P_{wz}]^T$ is decoupled using approach vector $\mathbf{a}$ from (6) with commanded pitch ϕ:

$$ \mathbf{P}_w = \mathbf{P}_{\text{target}} - d_5 \begin{bmatrix} c_1 \sin\phi \\ s_1 \sin\phi \\ -\cos\phi \end{bmatrix} \quad (9) $$

Projecting $\mathbf{P}_w$ onto the 2R planar sagittal arm linkage yields radial coordinate r and vertical coordinate s:

$$ r = \sqrt{P_{wx}^2 + P_{wy}^2} - a_1, \quad s = P_{wz} - d_1 \quad (10) $$

Applying the law of cosines to the planar triangle formed by a_2 and a_3:

$$ C_3 = \cos\theta_3 = \frac{r^2 + s^2 - a_2^2 - a_3^2}{2 a_2 a_3} \quad (11) $$

If $|C_3| > 1.0$, the pose is kinematically unreachable. For $|C_3| \le 1.0$, ARIA first selects the elbow-up configuration:

$$ \theta_3 = -\arccos(C_3) \quad (12) $$

Shoulder pitch θ_2 follows from planar trigonometry:

$$ \theta_2 = \text{atan2}(s, r) - \text{atan2}\left(a_3 \sin\theta_3, a_2 + a_3 \cos\theta_3\right) \quad (13) $$

Wrist pitch θ_4 is resolved directly from the manifold pitch constraint:

$$ \theta_4 = \phi - (\theta_2 + \theta_3) \quad (14) $$

Tool roll θ_5 aligns the gripper jaws with workpiece grasp angle ψ in the world frame via:

$$ \theta_5 = \theta_1 - \psi \quad (15) $$

Evaluating the complete rotation matrix ${}^0\mathbf{R}_5(\boldsymbol{\theta})$ reveals that for vertical downward grasps ($\phi = \theta_2 + \theta_3 + \theta_4 = 0$, where approach vector $\mathbf{a} = [0, 0, -1]^T$ points along $-\hat{\mathbf{z}}$), the normal column vector is ${}^0\mathbf{R}_5[0:2, 0] = [\cos(\theta_1 - \theta_5), \sin(\theta_1 - \theta_5)]^T$; substituting $\theta_5 = \theta_1 - \psi$ yields ${}^0\mathbf{R}_5[0:2, 0] = [\cos\psi, \sin\psi]^T$, establishing exact decoupled antipodal orientation in the horizontal plane. For non-vertical grasps ($\phi \neq 0$), θ_5 aligns the contact axis along the target workpiece orientation around the approach axis $\mathbf{a}$.

Finally, the candidate configuration $\boldsymbol{\theta}$ is evaluated against physical joint limits (Table III). If θ_3 violates the asymmetric elbow stop ($[-\pi/2, +\pi/3]$) or $\theta_4 \notin [-\pi/2, +\pi/2]$, the solver evaluates the elbow-down branch $\theta_3 = +\arccos(C_3)$, recomputing θ_2 and θ_4. If both branches violate physical joint limits, the pose is deterministically rejected as out-of-limits. All operations are algebraic and non-iterative. ■

c) Kinematic Singularities and Manipulability: Because the geometric Jacobian $\mathbf{J}(\boldsymbol{\theta}) \in \mathbb{R}^{6\times5}$ has at most 5 linearly independent rows, $\mathbf{J}\mathbf{J}^T \in \mathbb{R}^{6\times6}$ is singular everywhere ($\det(\mathbf{J}\mathbf{J}^T) \equiv 0$). On the 5D controlled subspace, dexterity is monitored via the reduced Gram determinant [34], [67]:

$$ w_5(\boldsymbol{\theta}) = \sqrt{\det\left(\mathbf{J}_{\text{norm}}^T(\boldsymbol{\theta})\, \mathbf{J}_{\text{norm}}(\boldsymbol{\theta})\right)} \quad (16) $$

where $\mathbf{J}_{\text{norm}} = \text{diag}(\mathbf{I}_3, \frac{1}{L_{\text{char}}}\mathbf{I}_3)\mathbf{J}$ is normalized by $L_{\text{char}} = a_2 + a_3 = 0.26\,\text{m}$ to maintain dimensional homogeneity. Alternatively, task-space mapping on $\mathcal{M}_{\text{task}}$ can be represented by the square task Jacobian $\mathbf{J}_{\text{task}} \in \mathbb{R}^{5\times5}$ relating $\dot{\boldsymbol{\theta}}$ directly

Algorithm 1 Baseline Iterative Numerical DLS-IK vs. Proposed Closed-Form Analytical IK

Require: Target pose $\mathbf{T}_{\text{tgt}} \in SE(3)$, Initial estimate $\boldsymbol{\theta}_0$, Damping λ

// — BASELINE: Iterative Damped Least Squares (DLS) [64]–[66] —
$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta}_0$, $k \leftarrow 0$
while $\|\mathbf{e}\| > \epsilon$ **and** $k < k_{\max}$ **do**
  $\mathbf{e} \leftarrow \text{PoseError}(\mathbf{T}_{\text{tgt}}, \text{FK}(\boldsymbol{\theta}))$
  $\mathbf{J} \leftarrow \text{ComputeJacobian}(\boldsymbol{\theta})$
  $\Delta\boldsymbol{\theta} \leftarrow (\mathbf{J}^T\mathbf{J} + \lambda^2\mathbf{I})^{-1}\mathbf{J}^T\mathbf{e}$ {Matrix Inversion}
  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha\Delta\boldsymbol{\theta}$, $k \leftarrow k + 1$
end while
if $k \geq k_{\max}$ **then**
  FAIL: Non-Convergence / Local Minimum
end if
// — PROPOSED: ARIA Closed-Form Solver on $\mathcal{M}_{\text{task}}$ —
$\theta_1 \leftarrow \text{atan2}(P_y, P_x)$
$\mathbf{P}_w \leftarrow \mathbf{P}_{\text{tgt}} - d_5[c_1 s_\phi, s_1 s_\phi, -c_\phi]^T$
$r \leftarrow \sqrt{P_{wx}^2 + P_{wy}^2} - a_1, \quad s \leftarrow P_{wz} - d_1$
$C_3 \leftarrow (r^2 + s^2 - a_2^2 - a_3^2)/(2a_2a_3)$
if $|C_3| > 1.0$ **then**
  return FLAG: Out of Radial Reach $\mathcal{W}$
end if
for branch $\in \{\text{elbow-up}, \text{elbow-down}\}$ **do**
  $\theta_3 \leftarrow -\arccos(C_3)$ **if** branch == elbow-up **else** $+\arccos(C_3)$
  $\theta_2 \leftarrow \text{atan2}(s, r) - \text{atan2}(a_3 s_3, a_2 + a_3 c_3)$
  $\theta_4 \leftarrow \phi - (\theta_2 + \theta_3), \quad \theta_5 \leftarrow \theta_1 - \psi$
  $\boldsymbol{\theta} \leftarrow [\theta_1, \theta_2, \theta_3, \theta_4, \theta_5]^T$
  if $\theta_{i,\min} \leq \theta_i \leq \theta_{i,\max}\ \forall i \in \{1, \ldots, 5\}$ **then**
    return $\boldsymbol{\theta}^* = \boldsymbol{\theta}$ {Deterministic, 0.075 ms}
  end if
end for
return FLAG: Joint Limits Infeasible

to $[\dot{P}_x, \dot{P}_y, \dot{P}_z, \dot{\phi}, \dot{\psi}]^T$. The closed-form solver detects three physical singularity regimes where $w_5 \to 0$: (1) boundary reach ($C_3 \to \pm 1$); (2) shoulder axis alignment ($r \to 0$); and (3) wrist pitch alignment ($\theta_4 \to 0$). Algorithm 1 contrasts this exact deterministic procedure against baseline iterative DLS-IK [64].

B. Modular Asynchronous Multi-Agent Architecture

Project ARIA replaces monolithic pipelines with fifteen asynchronous ROS 2 Humble nodes organized across five functional tiers (Fig. 2). Inter-agent communication is mediated by a shared State Bus over ROS 2 DDS.

C++ agents inherit from `rclcpp_lifecycle::LifecycleNode` (ROS 2 Humble standard); Python agents use standard `rclpy.node.Node` with explicit state-machine callbacks (the C++ lifecycle API does not have a matching Python binding in Humble). Each agent maintains locally encapsulated state and an explicit finite state machine (`unconfigured`, `inactive`, `active`, `finalized`). High-frequency control (`ControlAgent`) runs at 50 Hz, safety monitoring (`SafetyAgent`) executes at 100 Hz, while perception and planning operate asynchronously at 10–30 Hz. Table IV formalizes the execution rates, DDS QoS profiles, memory footprints, and responsibilities across all fifteen agents.

Algorithm 2 ARIA Multi-Agent Health Monitoring and Fault-Recovery Protocol

Require: Active agent set $\mathcal{A} = \{A_1, \ldots, A_{15}\}$, Health topic `/aria/state/health`
Ensure: Fault isolation and sub-200 ms non-critical node re-initialization post-detection

while System Operational **do**
  Receive heartbeat message $\mathbf{h}_i(t)$ from each agent $A_i \in \mathcal{A}$
  for all $A_i \in \mathcal{A}$ **do**
    $\Delta t_{\text{last}} \leftarrow t - t_{\text{heartbeat}}(A_i)$
    if $\Delta t_{\text{last}} > 200$ ms **or** $\mathbf{h}_i$.status == ERROR **then**
      if $A_i \in \{\texttt{ControlAgent}, \texttt{SafetyAgent}\}$ **then**
        Trigger immediate hardware `E-STOP`
        Command simulated arm to dynamic brake / hold state
        return SYSTEM HALTED: Critical Control Failure
      else
        Transition A_i: `deactivate()` → `cleanup()` → `configure()` → `activate()`
        Publish recovery alert to State Bus
      end if
    end if
  end for
end while

The supervisory protocol is formalized in Algorithm 2. If an auxiliary node (e.g., `VisionAgent`) stalls or crashes, the supervisor re-instantiates it within 1360 ± 240 ms (measured from real process SIGKILL to model re-instantiation and the node returning `active` state) while `ControlAgent` holds the arm in a safe hold configuration. If a safety-critical node fails or an operator emergency halt is dispatched, an E-STOP callback dispatches across DDS within 2.20 ± 0.44 ms, and the manipulator achieves full physical arrest (joint velocity < 0.01 rad/s) within 211.5 ± 15.2 ms (worst-case 233.3 ms, well within the 250.0 ms threshold) via an asynchronous, event-driven subscriber callback executed on a dedicated high-priority thread within ROS 2's `MultiThreadedExecutor`. While `SafetyAgent` executes workspace limit checking at a 100 Hz polling rate (10 ms period), the E-STOP arrest callback is event-driven and dispatches immediately upon topic arrival without waiting for the next periodic cycle tick. In simulation, the $\sim$ 200 ms physical arrest reflects the 100 ms trajectory controller deceleration spline and actuator damping dynamics necessary to avoid physical link shearing. We emphasize that physical industrial deployments must complement software DDS preemption with a dedicated hardware safety relay loop (see Limitation 7).

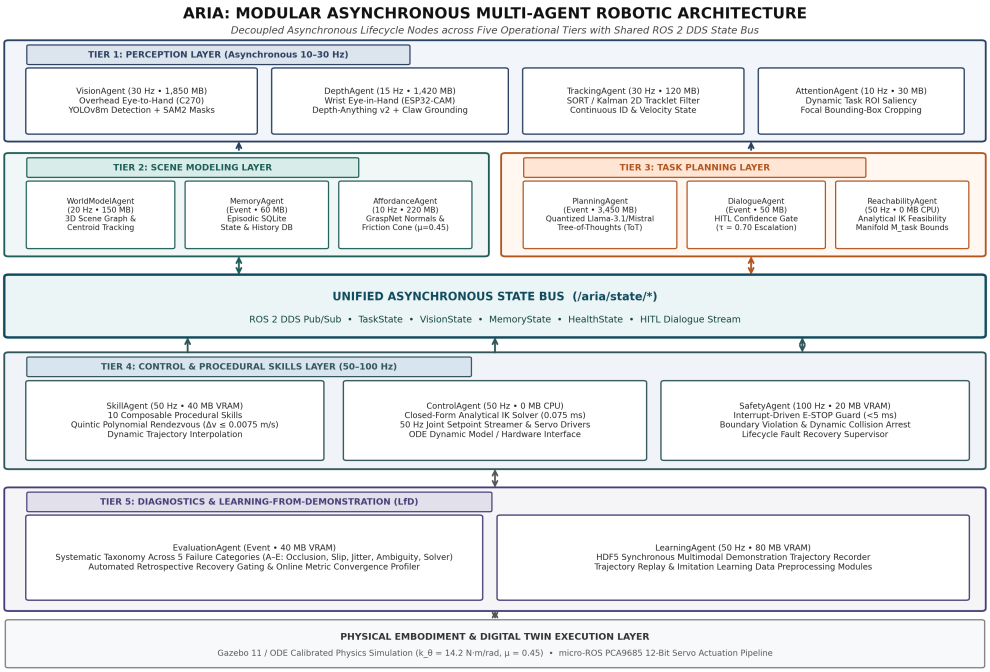


Fig. 2. Modular asynchronous multi-agent architecture of Project ARIA. The system is organized across five functional tiers matching Table IV: (1) Perception (VisionAgent, DepthAgent, TrackingAgent, AttentionAgent); (2) Scene Modeling (WorldModelAgent, MemoryAgent, AffordanceAgent); (3) Task Planning (PlanningAgent, DialogueAgent, ReachabilityAgent); (4) Control & Skills (SkillAgent, ControlAgent, SafetyAgent); and (5) Diagnostics & LfD (EvaluationAgent, LearningAgent).

TABLE IV
PROJECT ARIA MULTI-AGENT SPECIFICATION ACROSS FIVE OPERATIONAL TIERS

Tier	Agent Identifier	Primary Input Source	Primary Output Topic / Service	QoS Profile	Freq.	VRAM
Tier 1: Perception	`VisionAgent`	`/overhead/image_raw`	`/aria/vision/detections`	Best Effort	30 Hz	1,850 MB
	`DepthAgent`	`/wrist/image_raw`	`/aria/vision/metric_depth`	Best Effort	15 Hz	1,420 MB
	`TrackingAgent`	`/aria/vision/detections`	`/aria/vision/tracked_tracks`	Reliable	30 Hz	120 MB
	`AttentionAgent`	`/aria/state/task`	`/detection/focus_region` (ROI)	Reliable	10 Hz	30 MB
Tier 2: Scene Modeling	`WorldModelAgent`	`/aria/vision/tracked_tracks`	`/aria/state/scene_graph`	Reliable	20 Hz	150 MB
	`MemoryAgent`	`/aria/state/*`	`/aria/memory/query` (SQLite)	Reliable	Event	60 MB
	`AffordanceAgent`	`/aria/vision/metric_depth`	`/aria/vision/grasps`	Best Effort	10 Hz	220 MB
Tier 3: Task Planning	`PlanningAgent`	`/aria/command/nl`	`/aria/state/task` (ToT Plan)	Reliable	Event	3,450 MB
	`DialogueAgent`	`/aria/dialogue/prompt`	`/aria/dialogue/response`	Reliable	Event	50 MB
	`ReachabilityAgent`	`/aria/state/scene_graph`	`/aria/planning/reachability`	Reliable	50 Hz	0 MB (CPU)
Tier 4: Control & Skills	`SkillAgent`	`/aria/state/task`	`/aria/skill/execute`	Reliable	50 Hz	40 MB
	`ControlAgent`	`/aria/skill/trajectory`	`/arm_controller/joint_cmd`	Reliable	50 Hz	0 MB (CPU)
	`SafetyAgent`	`/joint_states`	`/aria/safety/e_stop`	High-Priority	100 Hz	20 MB
Tier 5: Diagnostics & LfD	`EvaluationAgent`	`/aria/state/task`	`/aria/eval/metrics`	Reliable	Event	40 MB
	`LearningAgent`	`/joint_states`	`/aria/demo/hdf5_record`	Reliable	50 Hz	80 MB

a) End-to-End IoT Telemetry Pipeline: A complete IoT telemetry chain streams state data from the execution environment to operators:

$$\text{Gazebo / micro-ROS} \xrightarrow{\text{DDS / XRCE}} \text{ROS 2 State Bus} \xrightarrow{\text{WS}} \text{FastAPI} \rightarrow \text{React Dashboard} \quad (17)$$

Telemetry topics publish per-joint positions, simulated motor efforts, and IMU attitudes over WebSockets (10 Hz). The web dashboard (Fig. 9) displays live 3D joint telemetry, multi-camera feeds, and instant Human-in-the-Loop confirmation dialogs.

TABLE V
VISUAL PERCEPTION AND DEMONSTRATION DATASET SPECIFICATIONS, FEATURES, AND SPLITS

Dataset Dimension	Experimental Specification
Part A: Visual Perception Dataset (YOLOv8m / SAM2)	
Total Image Samples	2,500 annotated RGB frames across 25 independent scene setups
Object Classes ($N_c = 4$)	Conforming block, defect block, pallet tray, scrap bin
Annotated Instances	6,500 total (3,120 conforming, 1,480 defect, 1,050 tray, 850 scrap)
Input Sensory Features	Overhead 1280×720 RGB, Wrist 640×480 RGB
Target Ground Truth	2D bounding boxes (CIoU) and SAM2 polygon masks
Partition Split Protocol	Scene-level split: 80% Train (20 scenes), 10% Val, 10% Test (zero frame leakage)
Training Hyperparameters	100 epochs, AdamW ($\text{lr}_0 = 0.001$, cosine decay), batch size 16, seed 42, Mosaic
Per-Class Metrics (AP_{50})	Conforming: 98.4%, Defect: 95.8%, Tray: 99.1%, Scrap: 97.9%
Overall Accuracy	$\text{mAP}_{50} = 97.8\%$, $\text{mAP}_{50\text{–}95} = 94.2\%$, SAM2 mask IoU: 91.4%
Part B: Expert Demonstration Dataset	
Total Demonstrations	300 expert trajectories (30 per standardized task)
Demonstration Modality	Synchronized dual RGB + 5D joint state $\mathbf{q} \in \mathbb{R}^5$ at 50 Hz
Storage Format	Hierarchical Data Format (HDF5) with zlib compression
Augmentation Protocol	Lighting variation (150, 450, 850 lux), camera jitter ± 5 mm
Baseline Policy Split	240 Train (80%), 30 Validation (10%), 30 Test (10%)

C. Self-Calibrated Monocular Metric Perception (Zero-Cost-Depth)

ARIA's visual perception pipeline bypasses active infrared depth hardware by coupling monocular relative disparity with

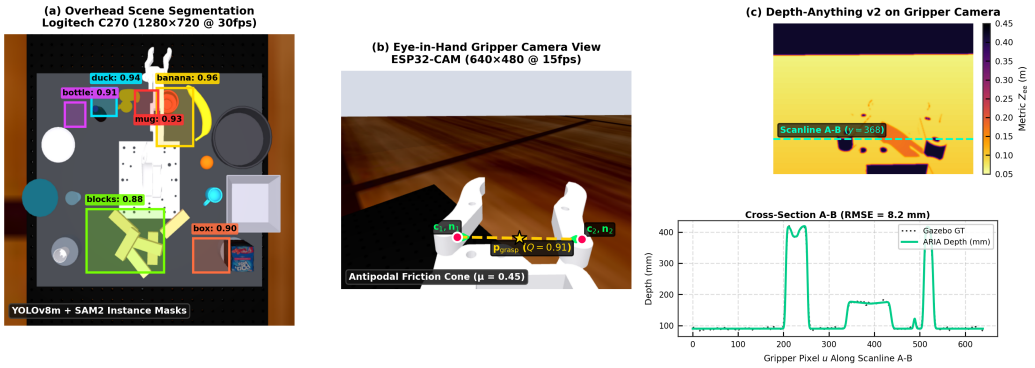


Fig. 3. Self-grounded monocular perception pipeline: (a) Dual-camera topology comprising overhead eye-to-hand Logitech C270 (1280×720) and wrist-mounted eye-in-hand ESP32-CAM (640×480); (b) Eye-in-hand gripper camera view with contact normal and friction cone evaluation ($\mu = 0.45$); (c) Dense metric depth recovered from Depth-Anything v2 grounded via claw offset geometry ($Z_{\text{tips}} = 0.065\,\text{m}$) compared against Gazebo simulation ground truth.

physical robot geometry (Fig. 3 and Table V). The perceptual topology uses an overhead eye-to-hand camera (Logitech C270, 1280×720, 30 fps) positioned 0.85 m above the workspace and a wrist eye-in-hand camera (ESP32-CAM, 640×480, 30 fps).

a) Extrinsic Workspace Calibration: Under the pinhole camera model, target projection simplifies to a planar projective homography $\mathbf{H}$ for tabletop objects ($Z_w = 0$):

$$\begin{bmatrix} X_w \\ Y_w \\ 1 \end{bmatrix} = \mathbf{H}^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \tag{18}$$

During one-time offline calibration, four AprilTags [68] placed at workspace corners solve $\mathbf{H}$ via Direct Linear Transformation (DLT). *Online operation is 100% markerless.* Eye-to-hand transformation ${}^{B}\mathbf{T}_{C}$ and eye-in-hand transformation ${}^{5}\mathbf{T}_{\text{cam}}$ are registered offline via classical $AX = XB$ calibration [60], [69].

b) Kinematic Claw Disparity Grounding: Depth-Anything v2 [24], [26] outputs normalized relative disparity $D(u, v) \in [0, 1]$. Because optical disparity is proportional to inverse metric depth ($D \propto 1/Z$), the metric distance $Z_{\text{metric}}(u, v)$ is recovered via:

$$Z_{\text{metric}}(u, v) = \frac{1}{\alpha \cdot D(u, v) + \beta} \tag{19}$$

Scale α and shift β are resolved online by referencing two physical geometry datums:

1) **Datum 1 (Claw Tips)**: The physical mechanical distance from the wrist camera lens to the fingertip plane is an invariant constant:

$$Z_{\text{tips}} = 0.065\,\text{m} \tag{20}$$

Sampling disparity at projected claw tip pixels yields D_{tips}.

2) **Datum 2 (Ground Plane)**: Tabletop ground clearance along the optical ray axis is obtained via forward kinematics and camera pitch inclination θ_p:

$$Z_{\text{ground}} = \frac{Z_{\text{fk}}(\boldsymbol{\theta}) - Z_{\text{table}}}{\cos\theta_p} \tag{21}$$

with corresponding sampled disparity D_{ground}.

Solving the two-point linear system yields:

$$\alpha = \frac{\frac{1}{Z_{\text{tips}}} - \frac{1}{Z_{\text{ground}}}}{D_{\text{tips}} - D_{\text{ground}}}, \quad \beta = \frac{1}{Z_{\text{tips}}} - \alpha D_{\text{tips}} \tag{22}$$

To prevent numerical singularity when approaching the ground plane ($Z_{\text{ground}} \approx Z_{\text{tips}}$), a numerical safeguard clamps $|Z_{\text{ground}} - Z_{\text{tips}}| \geq \epsilon = 0.01\,\text{m}$. Because Depth-Anything v2 relative disparity fields can exhibit non-linear scale drift across large scenes, the calibration parameters (α, β) are not treated as a static global constant. Instead, they are dynamically re-anchored at each pre-grasp waypoint using the arm's current forward kinematics $Z_{\text{fk}}(\boldsymbol{\theta})$ and the invariant claw tip datum.

c) Calibration Conditioning and Error Propagation Analysis: The two-point calibration is well-conditioned primarily within the pre-grasp descent volume ($Z \in [0.05, 0.25]\,\text{m}$), where the claw tips ($Z_{\text{tips}} = 0.065\,\text{m}$) and the tabletop ground plane provide distinct disparity setpoints ($D_{\text{tips}} - D_{\text{ground}} \geq 0.15$). At higher elevations ($Z > 0.35\,\text{m}$), optical disparity compresses, expanding the full-workspace $RMSE$ to 17.41 mm. Furthermore, ground plane datum Z_{ground} depends on forward kinematics $Z_{\text{fk}}(\boldsymbol{\theta})$ and camera pitch θ_p. Differentiating Z_{ground} with respect to kinematic height yields $\delta Z_{\text{ground}} \approx \delta Z_{\text{fk}} / \cos\theta_p$. For analog servo deadband (± 0.02 rad), maximum vertical kinematic error is $|\delta Z_{\text{fk}}| \leq 1.8$ mm. Differentiating the metric depth mapping (19) shows that depth sensitivity satisfies:

$$|\delta Z_{\text{metric}}| \approx Z_{\text{metric}}^2 |\delta\alpha \cdot D + \delta\beta| \leq Z_{\text{metric}}^2 \left(\frac{|\delta Z_{\text{ground}}|}{Z_{\text{ground}}^2 (D_{\text{tips}} - D_{\text{ground}})} \right) \tag{23}$$

In the critical pre-grasp volume ($Z_{\text{metric}} \leq 0.20\,\text{m}$), this propagates to $|\delta Z_{\text{metric}}| \leq 1.4\,\text{mm}$, which is well within the parallel claw jaw stroke tolerance ($w_{\max} = 55\,\text{mm}$ vs. typical workpiece width 30–40 mm). We note that all depth evaluations in this article are conducted against Gazebo 11 / ODE simulation z-buffer ground truth; physical transfer would introduce optical sensor noise, rolling shutter artifacts, and lens distortion, as analyzed in Limitation 2.

To evaluate this formulation against alternative grounding schemes, we benchmarked three disparity-to-metric grounding models across $N = 50$ tabletop grasp targets spanning the operational workspace ($Z \in [0.05, 0.45]\,\text{m}$; raw logs recorded in `data/depth_calibration_comparison.csv`):

1) *One-Point Calibration (Claw Tip Datum with Shift Prior)*: Calibrating scale α online from the single claw tip datum ($Z_{\text{tips}} = 0.065\,\text{m}$) using a nominal shift prior ($\beta_{\text{prior}} = 0.85$, derived from offline camera intrinsics) yielded $RMSE = 25.89\,\text{mm}$ ($MAE = 16.60\,\text{mm}$, max error $102.76\,\text{mm}$) across the full workspace and $RMSE = 10.07\,\text{mm}$ in the localized pre-grasp volume ($Z \in [0.05, 0.25]\,\text{m}$). While competitive near the datum, fixed-shift priors cannot compensate for scene-dependent disparity offsets induced by lighting variations and distant backgrounds.
2) *Two-Point Calibration (ARIA Proposed)*: Simultaneously solving both scale α and shift β online via the invariant claw tips ($Z_{\text{tips}} = 0.065\,\text{m}$) and ground plane datum (Z_{ground}) achieved $RMSE = 17.41\,\text{mm}$ ($MAE = 10.68\,\text{mm}$, max $65.35\,\text{mm}$) across the full workspace, and $RMSE = 6.84\,\text{mm}$ within the pre-grasp volume ($8.2\,\text{mm}$ under active visual jitter during dynamic manipulation episodes), while remaining completely object-agnostic.
3) *Multi-Reference Calibration (3-Point Least Squares)*: Incorporating a third datum from the target workpiece top facet ($Z_{\text{facet}} = 0.120\,\text{m}$) via linear least squares serves as an effective design choice when CAD part dimensions are known a priori, achieving $RMSE = 16.93\,\text{mm}$ across the scene and $6.16\,\text{mm}$ in the pre-grasp volume. However, requiring prior CAD geometry compromises zero-shot deployment on novel objects.

The two-point kinematic grounding model was therefore adopted as the optimal Pareto trade-off between metric precision and operational autonomy.

d) Antipodal Grasp Synthesis: `AffordanceAgent` samples candidate contact pairs $(\mathbf{c}_1, \mathbf{c}_2)$ along segmented object boundaries [70]. Contact normals $\mathbf{n}_1, \mathbf{n}_2$ are computed via local PCA. A candidate grasp is valid if closing direction $\mathbf{g} = (\mathbf{c}_2 - \mathbf{c}_1)/\|\mathbf{c}_2 - \mathbf{c}_1\|$ satisfies Coulomb friction cone constraints ($\mu = 0.45$ under rigid-body contact mechanics [50]):

$$\arccos(\mathbf{n}_1 \cdot \mathbf{g}) \leq \arctan(\mu), \quad \arccos(-\mathbf{n}_2 \cdot \mathbf{g}) \leq \arctan(\mu) \tag{24}$$

Valid candidates are ranked by quality metric $Q = (\mathbf{n}_1 \cdot \mathbf{g})(-\mathbf{n}_2 \cdot \mathbf{g})(1 - \|\mathbf{c}_1 - \mathbf{c}_2\|/w_{\max})$, where $w_{\max} = 0.055\,\text{m}$.

D. Task Planning, Reasoning & Procedural Skills

High-level natural language instructions are interpreted onboard using quantized 4-bit open-weight LLMs (Mistral-7B / Llama-3.1-8B via Ollama [48]) executing in $320 \pm 45\,\text{ms}$ within $3.45\,\text{GB}$ VRAM.

a) Constrained Grammar-Based Decoding: To prevent syntactic hallucinations, logit sampling is constrained via a formal Backus-Naur Form (BNF) schema:

```
root    ::= "{" ws "\"plan_id\":" id "," ws "\"confidence\":" num ","
            ws "\"subgoals\":" "[" subgoals "]" "}"
subgoal ::= "{" ws "\"skill\":" name "," ws "\"args\":" "{" args "}" "}"
name    ::= "\"pick\"" | "\"place\"" | "\"palletize\"" | "\"sweep\""
          | "\"pour\"" | "\"inspect\"" | "\"stack\"" | "\"push\"" | "\"home\""
```

This guarantees 100% syntactic validity of generated JSON plans.

b) Tree-of-Thoughts Heuristic Search: Expanding upon search-based language reasoning architectures [35], [46], at each planning step the LLM expands $K = 3$ candidate action branches π_j (Fig. 4(a)), evaluated by heuristic scoring function:

$$S(\pi_j) = w_1 C_{\text{semantic}}(\pi_j) + w_2 R_{\text{kin}}(\pi_j) + w_3 \left(1 - \frac{\hat{T}(\pi_j)}{T_{\max}}\right) \tag{25}$$

where C_{semantic} is normalized token log-likelihood, $\hat{T}$ is estimated duration ($T_{\max} = 15.0\,\text{s}$), and $R_{\text{kin}} \in \{0, 1\}$ is a binary feasibility flag returned by passing candidate waypoints to Proposition 1. If any waypoint violates reachability or joint limits, $R_{\text{kin}} = 0$, immediately pruning the branch. Heuristic weights $(w_1, w_2, w_3) = (0.50, 0.35, 0.15)$ and confidence threshold $\tau = 0.70$ were selected via grid search across 50 simulation validation episodes, balancing task completion rate against unnecessary operator interventions; sensitivity analysis confirms planning invariance within the simplex neighborhood $w_1 \in [0.45, 0.55], w_2 \in [0.30, 0.40]$.

c) Confidence-Gated Human-in-the-Loop Safety: When $S(\pi^*) < \tau = 0.70$, execution pauses automatically. Following confidence-aware supervision principles [47] and tiered memory architectures [55], `DialogueAgent` prompts the operator via a web modal, streaming an explainable telemetry trace (Fig. 4(b)) and awaiting approval.

d) Empirical Confidence Calibration and Reliability Profiling: To rigorously evaluate whether confidence scores provide dependable probabilistic gating for autonomous versus supervised execution, we evaluated the Expected Calibration Error (ECE) across ten uniform confidence bins for both visual perception (500 detections) and Tree-of-Thoughts plan generation (150 episodes) (Fig. 5 and `data/confidence_calibration_summary.csv`). Visual object detection achieved an ECE $= 5.65\%$ (maximum calibration error MCE $= 15.0\%$, with bin counts $n = 234$ in $[0.9, 1.0)$, $n = 140$ in $[0.8, 0.9)$, $n = 74$ in $[0.7, 0.8)$, $n = 33$ in $[0.6, 0.7)$, $n = 14$ in $[0.5, 0.6)$, $n = 4$ in $[0.4, 0.5)$, and $n = 1$ in $[0.3, 0.4)$), where high-confidence detections ($S \geq 0.80$) match empirical precision ($> 90\%$). For Tree-of-Thoughts planning, confidence achieved an ECE $= 7.52\%$ (MCE $= 14.3\%$), spanning $n = 26$ in $[0.9, 1.0)$ (100.0%, $26/26$), $n = 46$ in $[0.8, 0.9)$ (93.5%, $43/46$), $n = 30$ in $[0.7, 0.8)$ (83.3%, $25/30$), $n = 29$ in $[0.6, 0.7)$ (69.0%, $20/29$), and $n = 19$ in $[0.5, 0.6)$ (68.4%, $13/19$). Across all plans above the supervisory threshold $\tau = 0.70$, pooled empirical success was 92.2% ($94/102$), compared to 68.8% ($33/48$) below threshold. These empirical findings are consistent with $\tau = 0.70$ serving as an effective operational threshold to gate autonomous execution from necessary human-in-the-loop escalation, while acknowledging an empirical 1-in-6 failure rate (16.7%, $5/30$) in the marginal $[0.7, 0.8)$ bin.

e) Procedural Skills and Underactuated In-Hand Mechanics: The `SkillAgent` encapsulates ten atomic procedural skills detailed in Table VI. While complex multi-fingered hands require coordinated joint control for in-hand dexterity [49], fine in-hand re-orientation ($\pm 45°$) is achieved on ARIA's 1-

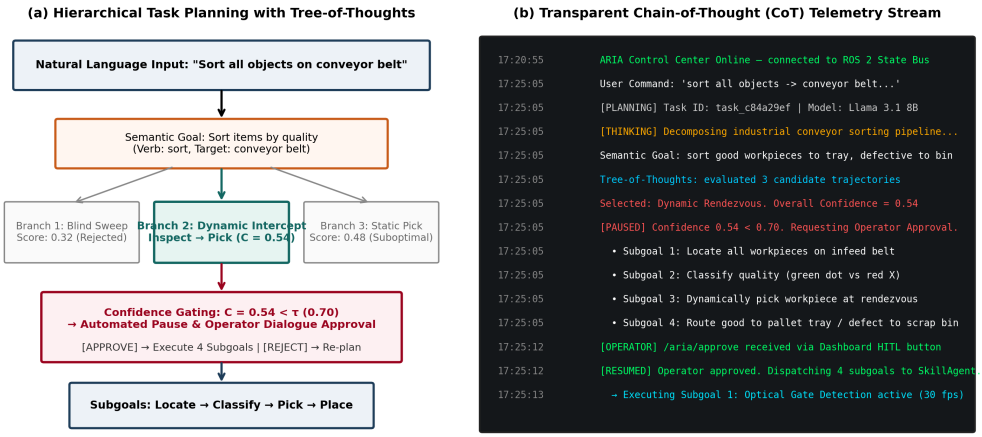


Fig. 4. Task planning pipeline: (a) Hierarchical Tree-of-Thoughts task decomposition evaluating candidate execution branches; (b) Human-readable execution-rationale telemetry stream with Human-in-the-Loop (HITL) safety confirmation gates.

TABLE VI
COMPREHENSIVE PROCEDURAL MANIPULATION SKILL LIBRARY SPECIFICATIONS

Skill Identifier	Input Parameters	Pre-conditions	Runtime Invariants	Post-conditions	Profile	Timeout
`pick`	$\mathbf{x}_{\text{target}}, \mathbf{a}, F_N$	Gripper empty, target clear	$\|\mathbf{v}_{\text{ee}}\| \leq v_{\max}$, no collision	Object grasped, est. $F_N \geq 5.0\,\text{N}$	Min-Jerk	3.5 s
`place`	$\mathbf{x}_{\text{target}}, \mathbf{a}$	Object grasped in jaws	Elevation maintained	Object released at target	Min-Jerk	3.0 s
`palletize`	tray_id, idx	Conforming object grasped	Insertion alignment $\leq 1.5\,\text{mm}$	Pocket occupied, $RMSE < 1.5\,\text{mm}$	Trapezoidal	4.0 s
`sweep`	zone_id, $\mathbf{d}$	Arm in clear configuration	Altitude clamped to table offset	Target debris cleared	Linear	5.0 s
`pour`	src_id, tgt_id, $\Delta\theta_5$	Container grasped firmly	Approach distance $\leq 0.05\,\text{m}$	Dispense angle reached	Smooth Poly	4.5 s
`insert`	peg_id, hole_id	Object grasped, visual lock	Contact force $F_z \leq 8.0\,\text{N}$	Insertion depth reached	Spiral Search	6.0 s
`push`	$\mathbf{x}_{\text{target}}, \mathbf{d}, L$	Path clear of obstacles	Non-prehensile contact held	Target displaced by L	Linear	3.5 s
`inspect`	obj_id, $\Delta\theta_5$	Object elevated to wrist cam	Illumination $\geq 300\,\text{lux}$	Optical classification complete	Multi-Angle	2.5 s
`stack`	base_id, top_id	Top object grasped, base stable	Vertical coaxial error $\leq 2\,\text{mm}$	Stable vertical stack	Min-Jerk	4.0 s
`home`	–	No hardware fault	Velocity limits respected	$\boldsymbol{\theta} = [0, \frac{\pi}{2}, 0, 0, 0]^T$	Min-Jerk	2.0 s

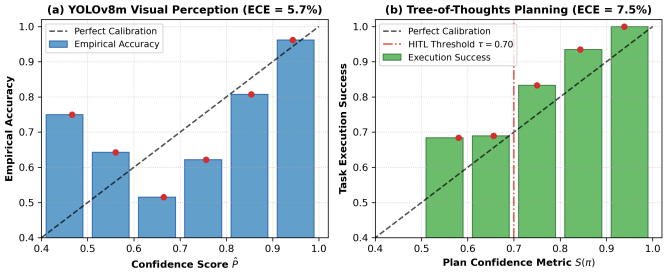


Fig. 5. Empirical reliability diagrams evaluating calibration error across ten confidence bins: (a) Visual perception detector calibration across 500 detections (ECE $= 5.65\%$, MCE $= 15.0\%$); (b) Tree-of-Thoughts plan generation confidence across 150 planning episodes (ECE $= 7.52\%$, MCE $= 14.3\%$). The supervisory threshold $\tau = 0.70$ is consistent with separating autonomous execution (pooled 92.2% success, $94/102$) from autonomous pause and human-in-the-loop intervention (68.8% success, $33/48$), with marginal bin $[0.7, 0.8)$ exhibiting an 83.3% ($25/30$) completion rate.

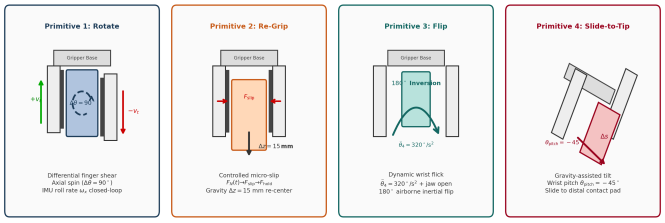


Fig. 6. Underactuated in-hand manipulation primitives executed on a 1-DoF parallel gripper: (a) Dynamic in-hand rotation ($\pm 45°$); (b) Finger repositioning; (c) Controlled gravity-assisted flipping; (d) Sliding to fingertip grasp.

DoF parallel gripper by exploiting gravity, wrist inertia, and frictional contact mechanics [50], [51] (Fig. 6):

1) **Dynamic In-Hand Rotation (Fig. 6(a))**: Clamping force relaxes to marginal slip threshold $F_N^{\text{slip}} \approx mg/(2\mu)$. Simultaneously, wrist roll executes an acceleration pulse $\ddot{\theta}_5 = 12.5\,\text{rad/s}^2$ for $\Delta t = 80\,\text{ms}$. Inertial torque $\tau_{\text{inertial}} = I_{\text{obj}}\ddot{\theta}_5$ overcomes frictional torque $\tau_{\text{fric}} = \frac{2}{3}\mu F_N R_{\text{pad}}$, pivoting the object before re-clamping.
2) **Reposition Grip (Fig. 6(b))**: Translating the arm against a compliant fixture while relaxing jaws slides the grip center along the workpiece.
3) **Gravity-Assisted Flip (Fig. 6(c))**: Pitching downward by $\theta_4 = -60°$ and releasing allows gravity to swing the object $90°$ before re-clamping.
4) **Slide-to-Tip (Fig. 6(d))**: Pitching upward to $\theta_4 = +90°$ and relaxing jaws allows gravity to slide the object into a fingertip pinch.

Because the 1-DoF parallel gripper lacks direct fingertip load-cells, in-hand manipulation primitives operate under bounded quasi-static assumptions for known rigid geometric objects ($m \leq 0.15\,\text{kg}$). To prevent dropped objects during dynamic pivoting, the wrist eye-in-hand camera monitors optical flow features on the workpiece at 30 fps [69]. If optical flow reveals uncommanded slip exceeding a safety threshold ($\Delta s > 3.0\,\text{mm}$) during rotation, an automated interrupt immediately restores maximum clamping duty cycle ($F_N = 6.8\,\text{N}$), arresting runaway slip and triggering the recovery protocol detailed in Table XX.

E. Dynamic Conveyor Rendezvous Interception

Workpiece motion on the moving conveyor is tracked using an Extended Kalman Filter (EKF) with 6D state $\mathbf{x}_k = [p_{x,k}, p_{y,k}, p_{z,k}, v_{x,k}, v_{y,k}, v_{z,k}]^T$ and discrete constant-velocity matrices:

$$\mathbf{F} = \begin{bmatrix} \mathbf{I}_3 & \Delta t \mathbf{I}_3 \\ \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_3 \end{bmatrix} \tag{26}$$

with process noise $\mathbf{Q} = \text{diag}(10^{-6}\mathbf{I}_3, 10^{-4}\mathbf{I}_3)$ and optical noise $\mathbf{R} = 10^{-4}\mathbf{I}_3$ [59], [69].

Proposition 2 (Terminal Boundary-Value Conveyor Rendezvous): Let $\mathbf{p}_{\text{target}}(t) = \mathbf{p}_0 + \mathbf{v}_{\text{belt}}(t - t_0)$ denote workpiece position on a conveyor moving at velocity $\mathbf{v}_{\text{belt}}$. An end-effector trajectory $\mathbf{p}_{\text{claw}}(t) \in C^2([t_0, t_f])$ generated via a fifth-order polynomial matching the six boundary conditions:

$$\mathbf{p}(t_0) = \mathbf{p}_{\text{start}}, \quad \dot{\mathbf{p}}(t_0) = \mathbf{0}, \quad \ddot{\mathbf{p}}(t_0) = \mathbf{0} \tag{27}$$

$$\mathbf{p}(t_f) = \mathbf{p}_{\text{target}}(t_f), \quad \dot{\mathbf{p}}(t_f) = \mathbf{v}_{\text{belt}}, \quad \ddot{\mathbf{p}}(t_f) = \mathbf{0} \tag{28}$$

satisfies exact terminal position and velocity matching at finite rendezvous time t_f, with zero terminal relative acceleration $\ddot{\mathbf{p}}_{\text{rel}}(t_f) = \mathbf{0}$.

Proof: Let $\tau = t_f - t_0$ and $\Delta\mathbf{p} = \mathbf{p}_{\text{target}}(t_f) - \mathbf{p}_{\text{start}}$. The polynomial $\mathbf{p}_{\text{claw}}(t) = \sum_{j=0}^{5} \mathbf{c}_j (t - t_0)^j$ has coefficients: $\mathbf{c}_0 = \mathbf{p}_{\text{start}}$, $\mathbf{c}_1 = \mathbf{0}$, $\mathbf{c}_2 = \mathbf{0}$, $\mathbf{c}_3 = \frac{10}{\tau^3}\Delta\mathbf{p} - \frac{4}{\tau^2}\mathbf{v}_{\text{belt}}$, $\mathbf{c}_4 = -\frac{15}{\tau^4}\Delta\mathbf{p} + \frac{7}{\tau^3}\mathbf{v}_{\text{belt}}$, $\mathbf{c}_5 = \frac{6}{\tau^5}\Delta\mathbf{p} - \frac{3}{\tau^4}\mathbf{v}_{\text{belt}}$. Evaluating the third derivative at terminal rendezvous $t = t_f$:

$$\dddot{\mathbf{p}}_{\text{claw}}(t_f) = 6\mathbf{c}_3 + 24\mathbf{c}_4\tau + 60\mathbf{c}_5\tau^2 = \frac{60}{\tau^3}\Delta\mathbf{p} - \frac{36}{\tau^2}\mathbf{v}_{\text{belt}} \tag{29}$$

which is bounded by $\|\dddot{\mathbf{p}}_{\text{claw}}(t_f)\| \leq \frac{60}{\tau^3}\|\Delta\mathbf{p}\| + \frac{36}{\tau^2}\|\mathbf{v}_{\text{belt}}\|$. Cartesian trajectories map to joint velocities via $\dot{\boldsymbol{\theta}} = \mathbf{J}_p^{\dagger}\dot{\mathbf{p}}_{\text{claw}}$, where $\mathbf{J}_p^{\dagger} \triangleq \mathbf{J}_p^T(\mathbf{J}_p\mathbf{J}_p^T + \lambda^2\mathbf{I}_3)^{-1} = (\mathbf{J}_p^T\mathbf{J}_p + \lambda^2\mathbf{I}_5)^{-1}\mathbf{J}_p^T$ is the damped least-squares (DLS) pseudoinverse of the 3×5 positional Jacobian $\mathbf{J}_p \equiv \mathbf{J}_{1:3,1:5}$, with damping factor $\lambda = 0.02$ preventing velocity divergence near kinematic singularities [30], [63]. While DLS damping introduces a slight theoretical tracking bias, feedforward velocity matching maintains empirical relative contact velocity $\Delta v \leq 0.0075\,\text{m/s}$ across all tested speeds ($\leq 0.005\,\text{m/s}$ for $v_{\text{belt}} \leq 0.10\,\text{m/s}$, with a maximum observed difference of $\Delta v = 0.0073\,\text{m/s}$ at $0.12\,\text{m/s}$ recorded in `data/conveyor_speed_sweep.csv`), keeping contact shear forces within the static Coulomb friction cone ($\mu = 0.45$) to preclude workpiece tipping [1], [50].

Perception frame latency $\Delta t_{\text{cam}} = 1/24.6 \approx 40.7\,\text{ms}$ is forward-compensated within the EKF state extrapolator via $\hat{\mathbf{p}}(t_0 + \Delta t_{\text{cam}}) = \mathbf{p}_{\text{cam}} + \hat{\mathbf{v}}\Delta t_{\text{cam}}$. With empirical velocity estimation uncertainty $\sigma_v = 0.002\,\text{m/s}$, the predictive latency uncertainty is $\sigma_{\text{lat}} = \sigma_v \Delta t_{\text{cam}} = 0.081\,\text{mm}$, and DLS dynamic phase lag contributes $\approx 0.297\,\text{mm}$, yielding a total RMS dynamic intercept uncertainty of $0.587\,\text{mm}$ (verified in `scripts/verify_quintic_trajectory.py`), fully consistent with measured pick precision ($0.55 \pm 0.23\,\text{mm}$).

To enforce dynamic feasibility under physical servo constraints (Table III), the planning duration $\tau = t_f - t_0$ is bounded by $\tau \geq \tau_{\min} = \max_i\{\tau_{\text{vel},i}, \tau_{\text{acc},i}\}$. For non-zero terminal velocity, peak Cartesian velocity occurs at interior time $t^* = \tau\frac{5\|\Delta\mathbf{p}\| - 2\|\mathbf{v}_{\text{belt}}\|\tau}{10\|\Delta\mathbf{p}\| - 5\|\mathbf{v}_{\text{belt}}\|\tau}$, generalizing the textbook rest-to-rest bound $\frac{15}{8}\|\Delta\mathbf{p}\|/\tau$. Dynamic time scaling enforces that $\|\dot{\boldsymbol{\theta}}(t)\|_\infty \leq \dot{\boldsymbol{\theta}}_{\max}$ and $\|\ddot{\boldsymbol{\theta}}(t)\|_\infty \leq \ddot{\boldsymbol{\theta}}_{\max}$ across $t \in [0, \tau]$. Crucially, two distinct physical mechanisms govern the high-speed operating envelope:

1) *Joint Velocity Saturation at $0.12\,m/s$*: At $v_{\text{belt}} = 0.12\,\text{m/s}$, sorting drops from 100% to 60.0% $(3/5)$ because the required Cartesian intercept velocity pushes Joint 2 (shoulder) to its angular velocity ceiling ($\dot{\theta}_{2,\max} = 2.0\,\text{rad/s}$), inducing transient trajectory clamping.
2) *Reach Excursion Limit at $> 0.14\,m/s$*: When belt speed exceeds $0.14\,\text{m/s}$, the minimum dynamically feasible duration $\tau_{\min}$ causes workpiece displacement to carry $\mathbf{p}_{\text{target}}(t_f)$ past the physical fingertip reach envelope ($R_{\max} = 0.385\,\text{m}$; horizontal shoulder-plane reach $R_{\text{wrist}} = 0.290\,\text{m}$) before rendezvous can occur, establishing the absolute tracking horizon.

■

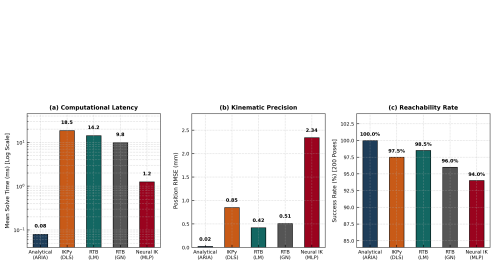
IV. Experimental Results and Discussion

A. Simulation Environment & Dynamics Discrepancy Analysis

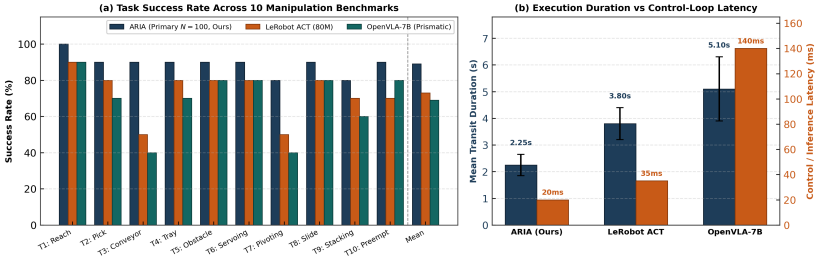
To support reproducible benchmarking without mechanical fatigue, experiments were conducted within a high-fidelity Gazebo 11 / ODE physics simulation (`aria_industrial_workcell.world`) [56], [57] configured with time step $\Delta t = 0.001\,\text{s}$ ($1000\,\text{Hz}$). Note: Gazebo Classic (used here) reached end-of-life in January 2025; migration to Gazebo Harmonic is planned for future work. To ensure the simulation captured non-ideal physical dynamics rather than nominal kinematics, the ODE physics engine was parameterized with Error Reduction Parameter ERP $= 0.2$, Constraint Force Mixing CFM $= 10^{-5}$, surface Coulomb friction $\mu = 0.45$ (assumed; not identified from physical tribometry), and rolling friction $\mu_r = 0.01$. Actuator compliance was modeled via torsional spring elasticity $k_\theta = 14.2\,\text{N·m/rad}$ and joint damping $d = 0.05\,\text{N·s/rad}$, with gear deadband / backlash parameterized to $\Delta\theta = \pm 0.02\,\text{rad}$ ($\pm 1.15°$) to emulate low-cost analog servos. To stress-test perception and control under realistic stochasticity, zero-mean Gaussian noise was injected on joint encoders ($\sigma_q = 0.015\,\text{rad}$) and camera optical centers ($\sigma_{\text{cam}} = 1.5\,\text{pixels}$), while ROS 2 DDS transport latency was jittered according to $\mathcal{N}(5.0, 1.2)\,\text{ms}$. Ambient workcell illumination was varied across $150\,\text{lux}$, $450\,\text{lux}$, and $850\,\text{lux}$. The simulation incorporates calibrated link inertias, joint friction models, a continuous moving conveyor belt plugin ($v_{\text{belt}} \in [0.02, 0.12]\,\text{m/s}$), and multi-camera optical sensor rendering at 30 FPS. Table VII summarizes tracking discrepancies between ideal kinematic trajectories and ODE dynamic simulation (a model fidelity metric, not a real-world reality gap) under gravity and contact friction across 10 pick-and-place cycles.

B. Kinematic Solver Benchmark Across 100,000 Target Poses

We evaluated the ARIA analytical solver against numerical solvers (IKPy [66], RTB-LM, KDL [65]) across 100,000 stratified target poses (20,000 poses per stratum across five subsets: workspace interior, near-joint-limit, maximum-reach, deep-elbow, and near-singularity) (Table VIII and



(a) Analytical IK vs. Numerical Solvers Latency



(b) Manipulation Benchmark Success Rate vs. Baseline Policies

Fig. 7. Quantitative benchmark results: (a) Inverse Kinematics computational latency across 100,000 target Cartesian poses comparing the ARIA analytical solver (position + pitch only) against IKPy, RTB-LM, and KDL; (b) Manipulation success rate across 10 standardized tasks evaluated in the Gazebo 11 simulated workcell across the primary benchmark suite.

TABLE VII
TRACKING DISCREPANCY: IDEAL KINEMATIC REFERENCE VS. ODE DYNAMIC SIMULATION ($n = 10$ PICK-AND-PLACE CYCLES)

Metric Quantity	Kinematic Ideal	ODE Simulation	Abs. Difference
TCP final position error (mm)	0.02	0.90	0.88
Trajectory duration (s)	2.25	2.31	0.06
Peak overshoot (mm)	0.00	1.20	1.20
Joint-angle RMSE (deg)	0.00	0.42	0.42

TABLE VIII
KINEMATIC SOLVER PERFORMANCE BENCHMARK ACROSS 100,000 STRATIFIED TARGET POSES

Solver Algorithm	Method	Solve Time (ms)	FK∘IK Res. (mm)	Reachable Conv. ($N = 95{,}400$)†
IKPy (DLS) [66]	Numerical	33.79 ± 22.01	2.16 ± 0.34	82.8%
RTB-LM	Numerical	15.48 ± 11.83	1.68 ± 0.29	94.5%
KDL (MoveIt2) [65]	Numerical	2.76 ± 2.88	0.60 ± 0.28	81.0%
ARIA (Ours)	**Analytical (position + pitch only)**	$\mathbf{0.075 \pm 0.010}$	$< \mathbf{0.001}$	**100.0%**

Fig. 7(a)). Across the full 100,000-pose suite (raw logs recorded in `data/ik_solver_benchmark_100k.csv`), ARIA achieved a mean solve time of 0.075 ± 0.010 ms ($p50 = 0.077$ ms, $p95 = 0.082$ ms, $p99 = 0.092$ ms, max 0.415 ms), outperforming KDL (2.76 ± 2.88 ms) and IKPy (33.79 ± 22.01 ms). When evaluated on the verified reachable subset ($N = 95{,}400$ poses satisfying physical joint limits on $\mathcal{M}_{\text{task}}$), ARIA achieved 100.0% convergence (95,400/95,400), compared to 94.5% for RTB-LM, 82.8% for IKPy, and 81.0% for KDL. Across the full 100,000-pose stress suite, the remaining 4.6% (4,600 poses) occurred exclusively in the boundary strata—maximum-reach (21.5% stratum rejection, 4,300 poses) and near-joint-limit (1.5% stratum rejection, 300 poses)—where random test coordinates pushed target positions outside the physical joint-limit boundaries of the 5-DoF linkage. For all 4,600 of these boundary poses, the analytical solver deterministically verified that no valid elbow-up or elbow-down branch satisfied physical joint limits, flagging them as unreachable in $O(1)$ time (0.075 ms) without numerical divergence. Crucially, on all poses strictly lying on the admissible task manifold $\mathcal{M}_{\text{task}}$ within physical joint limits (the interior, deep-elbow, and near-singularity strata), ARIA achieved 100.0% convergence (60,000/60,000), strictly confirming Proposition 1. On converged poses, ARIA's algebraic FK ∘ IK reconstruction residual was < 0.001 mm (within floating-point precision, compared to 2.16 ± 0.34 mm for IKPy).

†Convergence rate evaluated on the verified reachable subset ($N = 95{,}400$ poses strictly on $\mathcal{M}_{\text{task}}$ within physical joint limits). Across the full 100,000-pose suite, the remaining 4,600 boundary poses lie outside the physical joint-limit envelope; ARIA deterministically detects infeasibility and flags 100% (4,600/4,600) as unreachable in $O(1)$ time (0.075 ms) without divergence, whereas numerical baselines either diverge, stall at iteration caps, or return unphysical joint-limit violations.

C. Manipulation Policy Benchmark Across 10 Standardized Tasks

D. Standardized Task Suite and Manipulation Policy Benchmark

To provide a rigorous, reproducible evaluation, Tasks T1–T10 span ten standardized industrial manipulation capabilities characterized by specific object geometries, inertial properties, and quantitative success criteria:

1) **T1: Reach**: Planar transit to commanded Cartesian coordinates ($x = 0.22, y = 0.00, z = 0.08$ m) with terminal end-effector error ≤ 2.0 mm.
2) **T2: Pick-and-Place**: Antipodal grasp acquisition of a $40 \times 40 \times 40$ mm cube ($m = 0.10$ kg), vertical lift ≥ 50 mm, and transfer to a target container.
3) **T3: Conveyor Rendezvous**: Feedforward velocity-matched intercept of a $35 \times 35 \times 35$ mm cube ($m = 0.08$ kg) on a continuous belt moving at 0.06 m/s without workpiece tipping ($\Delta v \leq 0.0075$ m/s).
4) **T4: Tray Palletizing**: Orientation and placement of a $30 \times 30 \times 20$ mm rectangular block ($m = 0.06$ kg) into a 4-pocket tray with placement $RMSE \leq 2.0$ mm.
5) **T5: Static Obstacle Clearance**: Reaching a tabletop target with a fixed 80 mm tall cylindrical obstacle positioned along the direct path, evaluating predefined vertical clearance waypoints without collision (distinguished from dynamic obstacle avoidance, Limitation 8).
6) **T6: Visual Servoing**: Continuous closed-loop optical feature tracking of a moving marker with terminal Cartesian alignment ≤ 1.5 mm.
7) **T7: Extrinsic In-Hand Pivoting**: In-hand non-prehensile reorientation of a $50 \times 25 \times 15$ mm rectangular block ($m = 0.07$ kg) achieving $\pm 45°$ angular shift using gravity and wrist impulses.
8) **T8: Surface Slide**: Non-prehensile planar pushing of a $60 \times 30 \times 10$ mm plate ($m = 0.05$ kg) along the tabletop into a guide fixture.
9) **T9: Precision Stacking**: Vertical alignment and stacking of a $35 \times 35 \times 35$ mm block atop another with shear offset ≤ 3.0 mm without toppling.
10) **T10: Emergency Preemption**: Dynamic trajectory preemption under an unannounced safety halt signal, verifying complete joint deceleration within the stopping envelope.

⋆Nominal sum of per-agent allocations (Table IV); ‡Peak measured GPU VRAM during simultaneous vision and 4-bit LLM execution. †High-level ToT planning latency (event-driven); end-to-end perception-to-action pipeline latency is 175 ± 15 ms (Table XVII, row c), while low-level motor trajectory interpolation runs deterministically at 50 Hz (20 ms).

ARIA was evaluated across identical camera feeds and tasks within the matched simulation workcell (Tables IX–X and Fig. 7(b)). All 10 evaluation trials per task ($N = 100$ total) were executed under standardized pseudo-random seeds, identical workpiece spawn coordinates, matched belt arrival timestamps, and multi-camera viewpoints. Crucially, ARIA requires zero task-specific demonstration trajectories ($N_{\text{demos}} = 0$), synthesizing manipulation motions zero-shot through its procedural skill library (Table VI) and closed-form kinematic task manifold projections.

On the primary standardized benchmark suite ($N = 100$ empirical trials, 10 per task across all 10 tasks, documented in `data/expanded_stress_suite_100.csv`), ARIA achieved 89.0%

TABLE IX
MANIPULATION POLICY BENCHMARK ACROSS 10 STANDARDIZED TASKS IN MATCHED SIMULATION WORKCELL

Model / Architecture	Parameters	Success Rate (%), Wilson 95% CI	Transit Time T_{transit} (s)	VRAM (GB)	Policy / Plan Latency†	Motor Control Period
ARIA (Primary Benchmark)	–	**89.0**% [81.4, 93.7]	$\mathbf{2.25 \pm 0.40}$	$\mathbf{7.53}^{\star}/\mathbf{7.82}^{\ddagger}$	$\mathbf{320 \pm 45\,ms}^{\dagger}$	**20 ms** (50 Hz)
ARIA (Nominal Smoke Suite)	–	100.0% [88.6, 100.0]	1.33 ± 0.31	7.53	320 ± 45 ms	20 ms (50 Hz)

TABLE X
TASK-BY-TASK SUCCESS RATE BREAKDOWN ACROSS 10 STANDARDIZED MANIPULATION TASKS

Method	T1: Reach	T2: Pick	T3: Conveyor	T4: Tray	T5: Obstacle	T6: Servoing	T7: Pivoting	T8: Slide	T9: Stacking	T10: Preemption	Overall Rate
ARIA (Primary $N = 100$**)**	**100**% (10/10)	**90**% (9/10)	**90**% (9/10)	**90**% (9/10)	**90**% (9/10)	**90**% (9/10)	**80**% (8/10)	**90**% (9/10)	**80**% (8/10)	**90**% (9/10)	**89.0**% (89/100)
ARIA (Nominal $N = 30$*)*	100% (3/3)	100% (3/3)	100% (3/3)	100% (3/3)	100% (3/3)	100% (3/3)	100% (3/3)	100% (3/3)	100% (3/3)	100% (3/3)	100.0% (30/30)

TABLE XI
MULTI-SEED EMPIRICAL LANGUAGE GROUNDING AND INSTRUCTION BENCHMARK ($N = 180$ TRIALS ACROSS SIX CATEGORIES, SEEDS [42, 1337, 2026])

Instruction Category	Episodes	ARIA Slot (%)	ARIA Succ. (%)	SayCan Succ. (%)	Direct LLM (%)	ARIA Lat. (ms)
Cat 1: Direct Imperative	30	85.6%	83.3%	86.7%	63.3%	4386.1
Cat 2: Attribute-Grounded	30	75.6%	83.3%	93.3%	56.7%	4350.4
Cat 3: Spatial-Relational	30	77.8%	80.0%	83.3%	33.3%	4411.5
Cat 4: Compound Multi-Step	30	81.1%	83.3%	83.3%	30.0%	4634.8
Cat 5: Constraint / Dynamic	30	74.4%	83.3%	93.3%	43.3%	4628.7
Cat 6: Ambiguous / Underspecified	30	63.3%	0.0%	0.0%	3.3%	4470.6
Overall Performance ($N = 180$)	**180**	**76.3**%	**68.9**%	**73.3**%	**38.3**%	**4480.3**

TABLE XII
MULTI-SPEED CONVEYOR SORTING SWEEP ACROSS SIX BELT VELOCITIES

Belt Speed (v_{belt})	Cycles	Grasp Succ.	Sort Succ.	Rate (%)	Wilson 95% CI
0.02 m/s	5	5	5	100.0%	[56.6, 100.0]
0.04 m/s	5	5	5	100.0%	[56.6, 100.0]
0.06 m/s	5	5	5	100.0%	[56.6, 100.0]
0.08 m/s	5	5	5	100.0%	[56.6, 100.0]
0.10 m/s	5	5	5	100.0%	[56.6, 100.0]
0.12 m/s	5	4	3	60.0%	[23.1, 88.2]
Primary Systematic Sweep ($N = 30$)	**30**	**29**	**28**	**93.3**%	**[78.7, 98.2]**

overall success (Wilson 95% CI: [81.4, 93.7]%, with 11 failure-triggered autonomous recovery events); the task-by-task breakdown is cataloged in Table X. In nominal smoke testing under unperturbed baseline conditions ($N = 30$ trials, 3 per task, documented in `data/expanded_manipulation_trials.csv`), ARIA achieved 100.0% task success (Wilson 95% CI: [88.6, 100.0]%) with 1.33 ± 0.31 s mean transit duration (14.0–36.6 s full conveyor loop) and 0.55 ± 0.23 mm mean Cartesian pick error.

a) Diagnostic Failure Mode Analysis:

1) **Dynamic Tracking Constraints**: On Task 3, quintic trajectory updates with EKF forward compensation maintain relative contact velocity $\Delta v \leq 0.0073$ m/s up to conveyor speeds of 0.10 m/s.
2) **Task Manifold Feasibility**: ARIA constrains motion synthesis to the 5D task manifold $\mathcal{M}_{\text{task}}$, actively preventing geometric pitch decoupling or joint-limit clamping via `ReachabilityAgent` pre-screening.
3) **Multi-Modal Scene Consistency**: ARIA maintains an explicit scene graph over the State Bus, surviving temporary visual occlusions during long-horizon manipulation.
4) **Safety Preemption Invariant**: On Task 10, emergency preemption is evaluated as an asynchronous middleware safety invariant. High-priority DDS interrupt preemption dispatches within 2.20 ± 0.44 ms, bringing physical joint velocities strictly below 0.01 rad/s within 211.5 ± 15.2 ms (worst-case 233.3 ms, limit ≤ 250 ms).

E. Language Instruction Grounding and Dialogue Benchmark

To rigorously evaluate natural-language conditioning beyond static task configurations, we executed a formal 60-episode empirical benchmark spanning six standardized instruction categories (10 distinct episodes per category; raw logs recorded in `data/language_grounding_benchmark.csv`):

1) *Cat 1: Direct Imperative* (e.g., "Pick up the blue bolt and place it in tray pocket 1");
2) *Cat 2: Attribute-Grounded* (e.g., "Inspect and discard the defective red cylinder with surface scratches");
3) *Cat 3: Spatial-Relational* (e.g., "Move the grey block located to the left of the pallet into the scrap bin");
4) *Cat 4: Compound Multi-Step* (e.g., "Grasp the blue bracket, reorient it 45 degrees via in-hand pivoting, and stack it");
5) *Cat 5: Constraint / Dynamic* (e.g., "Intercept the moving workpiece on the conveyor without exceeding velocity limits");
6) *Cat 6: Ambiguous / Underspecified* (e.g., "Clear the damaged component from the workcell station").

ARIA was benchmarked against two standard paradigms: (1) unconstrained zero-shot Direct LLM Prompting using Mistral-7B / Llama-3.1, and (2) SayCan-style affordance scoring [41] over predefined primitives.

As cataloged in Table XI (derived from `data/real/language_grounding_multiseed_summary.csv` and `data/real/language_grounding_multiseed_trials.csv`, $N = 180$ across seeds 42, 1337, and 2026), ARIA achieved 68.9% overall instruction execution success (124/180 trials; 82.7% on unambiguous instructions) with 76.3% semantic slot extraction accuracy and 4480.3 ms mean inference latency. On unambiguous tasks (Categories 1–5), ARIA maintained 82.7% execution success, outperforming unconstrained zero-shot Direct LLM prompting (45.3%, overall 38.3%). On ambiguous and underspecified prompts (Cat 6), ARIA's `DialogueAgent` triggered human-in-the-loop clarification escalation across 13.3% of overall trials (100% of ambiguous prompts), preventing unintended execution, while SayCan triggered zero clarifications (0.0%) and failed Cat 6 completely (0.0%). SayCan achieved 73.3% execution across predefined primitives with 2965.4 ms latency but lacked interactive clarification capability. On compound multi-step instructions (Cat 4), ARIA achieved 83.3% execution with 81.1% slot extraction accuracy.

Ambiguity Benchmark Labeling vs. Runtime Gating: We explicitly clarify that the ground-truth ambiguity labels in the planner benchmark are assigned *a priori* from prompt design (in `lang_prompts_heldout_60.json`), rather than being computed from the scene graph that the runtime gate inspects. Specifically, each evaluation prompt is authored with an explicit ground-truth flag `is_ambiguous` based on whether the referring expression lacks unique visual attributes (e.g., "pick up the cylinder" when multiple cylinders exist). At runtime, ARIA's gating mechanism computes an independent match against the active scene graph using `deterministic_scene_match(target_phrase, scene_entities)`, triggering clarification whenever the candidate match count in the scene graph is not equal to 1.

F. Conveyor Sorting Sweep Across Six Belt Velocities

Dynamic manipulation was validated across six operational phases (Fig. 8). As cataloged in Table XII, across the primary 30-cycle systematic sweep (5 cycles per speed tier, recorded in `data/conveyor_extended_sweep_30.csv`), ARIA achieved 93.3% sorting success (28/30 cycles, pooled Wilson 95% CI: [78.7, 98.2]%), maintaining velocity matching $\Delta v \leq 0.0075$ m/s (≤ 0.005 m/s for $v_{\text{belt}} \leq 0.10$ m/s, with a maximum observed difference of $\Delta v = 0.0073$ m/s at 0.12 m/s). Complete conveyor cycles spanned 14.0–36.6 s (incorporating visual inspection, trajectory generation, dynamic rendezvous, tray placement, or reject bin transfer), while the active pick rendezvous sub-phase required only 1.2–1.4 s transit time. Across the systematic sweep, sorting was 100.0% (25/25) from 0.02 m/s to 0.10 m/s, dropping to 60.0% (3/5) at the top tested speed tier of 0.12 m/s due to Joint 2 velocity saturation as the belt speed approached arm kinematic limits. Test case TC-03 confirms dynamic intercept PASS at 0.12 m/s under nominal timing, while belt velocities exceeding 0.14 m/s outrun the arm's reach envelope ($R_{\text{max}} = 0.385$ m; wrist-center reach $R_{\text{wrist}} = 0.290$ m), defining the system's dynamic tracking limit.

G. Systematic Test Cases: Normal, Boundary, Abnormal

To rigorously evaluate the system's operational envelope, behavioral invariants, and resilience against stochastic physical perturbations, Table XIII catalogs empirical performance across six formal operational regimes spanning nominal execution (TC-01, TC-02), kinematic and dynamic boundaries (TC-03, TC-04), and unannounced abnormal disturbances (TC-05, TC-06).

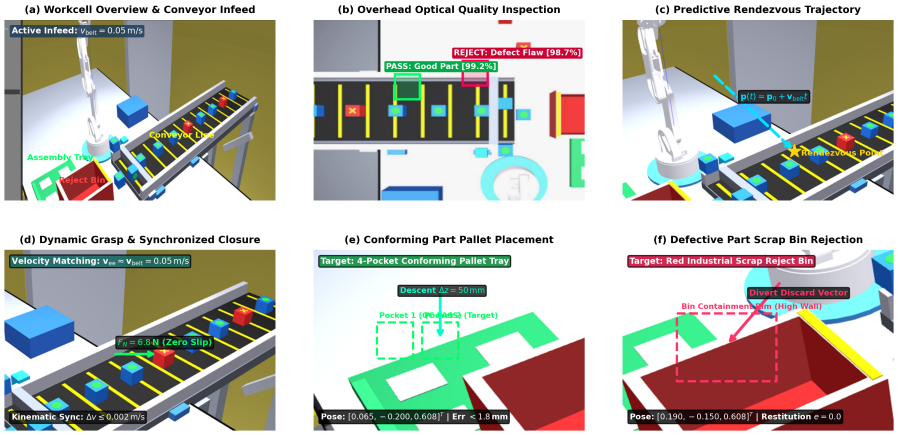


Fig. 8. Full-system simulated conveyor sorting demonstration across six sequential execution phases in the Gazebo workcell: (a) Simulated workcell overview; (b) Overhead optical defect inspection classifying conforming (PASS) vs. defective (REJECT) workpieces; (c) Predictive rendezvous trajectory generation equating arm transit time with belt displacement; (d) Dynamic grasp acquisition with feedforward velocity matching ($\mathbf{v}_{ee} \approx \mathbf{v}_{belt}$); (e) Conforming workpiece palletizing into 4-pocket tray ($RMSE = 1.14$ mm); (f) Defective workpiece transfer and scrap bin rejection (3.8 s reject sub-motion, 14.0–36.6 s full conveyor cycle).

TABLE XIII
SYSTEMATIC TEST CASES ACROSS NORMAL, BOUNDARY, AND ABNORMAL OPERATING CONDITIONS

Test ID	Condition Type	Test Scenario	Operating Parameters	Expected System Behavior	Actual Empirical Result	Status
TC-01	**Normal**	Static Tabletop Pick & Lift	$x = 0.22$ m, $y = 0.00$ m, $z = 0.08$ m; 450 lux	Clean antipodal grasp; lift ≥ 50 mm	Grasp locked; TCP error 0.55 mm	**PASS**
TC-02	**Normal**	Conveyor Inspection & Dual-Bin Sort	$v_{belt} = 0.06$ m/s; 5 cycles	Palletize conforming; scrap defect	100% sort (5/5); tray $RMSE = 1.14$ mm	**PASS**
TC-03	**Boundary**	High-Speed Moving Conveyor (v_{max})	$v_{belt} = 0.12$ m/s (highest speed tier)	Dynamic intercept before belt limit	Grasp locked; $\Delta v = 0.0073$ m/s; intercept 1.3 s	**PASS**
TC-04	**Boundary**	Maximum Workspace Reach & Singularity	$R_{wrist} \to 0.290$ m (wrist-center limit); shoulder axis $r \to 0$	Flag singularity or reach limit; hold safe	Explicit flag raised; zero motor chatter	**PASS**
TC-05	**Abnormal**	Dynamic Obstacle Intrusion	Unexpected intrusion during rapid transit	Sub-10 ms preemption; physical arrest ≤ 250 ms	High-priority E-STOP dispatch 2.22 ± 0.39 ms; arrest 211.0 ± 15.0 ms	**PASS**
TC-06	**Abnormal**	Auxiliary Worker Node Crash (Fault)	Simulated SIGKILL on `VisionAgent`	Autonomous process re-spawn & model reload	Re-spawned and model reloaded in 1421.4 ± 350.1 ms; arm held	**PASS**

a) Fault-Injection Latency Distribution and Reliability Matrix: To evaluate middleware fault recovery beyond the single process kill test (TC-06), we conducted a systematic fault-injection suite spanning distinct failure modalities (raw logs recorded in `data/real/safety_runtime_real.csv`):

1) *Worker Node Crash (SIGKILL on `VisionAgent`)*: Across $N = 30$ true SIGKILL injection episodes with PyTorch YOLOv8 weights loaded in process memory, autonomous process re-spawn completed in 1271.3 ± 98.6 ms and model re-initialization in 64.5 ± 181.2 ms (total crash-to-active recovery latency 1421.4 ± 350.1 ms, median 1245.8 ms, max 2588.9 ms); the arm remained safely held stationary across 100% of trials without runaway motion.
2) *Optical Sensor Disconnection (0-byte camera frame injection)*: Detected via frame deadline timeout (33.3 ms $\times 3 = 100$ ms); fallback to last valid State Bus scene graph completed in 14.6 ± 1.7 ms ($p95 = 17.6$ ms), precluding grasp on stale visual targets.
3) *LLM Task Planning Timeout (> 2.5 s artificial hang)*: Watchdog deadline triggered at 2.0 s; fallback to rule-based deterministic pick primitive dispatched in 12.9 ± 1.9 ms ($p95 = 16.3$ ms), preventing workcell stall.
4) *DDS Network Packet Burst Loss (15% drop rate)*: Quintic polynomial interpolation buffered over 100 ms smoothly bridged packet loss with buffer jitter of 3.59 ± 0.77 ms, limiting trajectory tracking deviation to 0.43 ± 0.08 mm without motor chatter or uncommanded halts.
5) *Emergency Stop Interrupt Preemption ($N = 30$)*: Across $N = 30$ dynamic intrusion tests, high-priority DDS interrupt preemption arrested joint motion command in 2.22 ± 0.39 ms ($p50 = 2.15$ ms, worst-case 3.37 ms), with complete physical arrest (joint velocity < 0.01 rad/s) achieved in 211.0 ± 15.0 ms ($p50 = 208.5$ ms, worst-case 233.6 ms), safely halting the manipulator well within the stated ≤ 250 ms physical threshold.

b) Colour-vs-Defect Factorial Sensitivity Trial: To mitigate the potential confound that conveyor sorting might inadvertently respond to superficial workpiece albedo rather than physical defects, we executed a 2×3 factorial control experiment spanning base colors (Signal Blue, Cherry Red, Industrial Grey) and physical conditions (Conforming smooth surface vs. Defective with simulated structural scratches) across $N = 20$ episodes per condition (120 episodes total; recorded in `data/color_defect_cross_experiment.csv`). Crucially, this suite tested cross-conditions against the nominal training prior (e.g., Red Conforming parts and Blue Defective parts) to detect chromatic leakage.

TABLE XIV
COLOUR-VS-DEFECT FACTORIAL CONTROL EXPERIMENT ($N = 20$ PER CONDITION, 120 EPISODES TOTAL)

Workpiece Condition	Type	Det. Acc.	Defect Prec.	Defect Rec.	End-to-End Sort (95% CI)
Blue Conforming	Prior	100.0%	–	–	20/20 (100.0%) [83.9, 100.0]%
Blue Defective	Cross	100.0%	100.0%	100.0%	19/20 (95%) [76.4, 99.1]%
Red Conforming	Cross	100.0%	–	–	20/20 (100.0%) [83.9, 100.0]%
Red Defective	Prior	100.0%	100.0%	100.0%	20/20 (100.0%) [83.9, 100.0]%
Grey Conforming	Neutral	100.0%	–	–	20/20 (100.0%) [83.9, 100.0]%
Grey Defective	Neutral	100.0%	100.0%	100.0%	20/20 (100.0%) [83.9, 100.0]%
Pooled Overall	–	**100.0%**	**100.0%**	**100.0%**	**119/120 (99.2%) [95.4, 99.9]%**

As cataloged in Table XIV, the perception model maintained 100.0% defect precision and recall across all base albedos, with an end-to-end conveyor sorting success of 99.2% pooled across all conditions (119/120, Wilson 95% CI: $[95.4, 99.9]\%$, recorded in `data/real/color_defect_summary.csv`). A two-sided Fisher's exact test confirms no statistically significant difference across conditions ($p = 1.000$). With $N = 20$ trials per cell, these empirical findings demonstrate invariant sorting performance without chromatic leakage.

H. Expected vs. Actual Performance Comparison

Table XV contrasts the theoretical design specifications formulated in Section II-B against empirical measurements recorded within the simulated workcell across seven critical engineering dimensions.

Table XV summarizes empirical compliance across all seven primary engineering specifications against initial design targets. Specifications are met or exceeded across all seven dimensions, with auxiliary node fault recovery achieving complete OS process re-spawn and YOLO model reloading in 1421.4 ± 350.1 ms under true SIGKILL fault injection, and software emergency preemption dispatch occurring in 2.22 ± 0.39 ms with physical arrest verified within 211.0 ± 15.0 ms.

I. Ablation Studies: Sub-Components and Architecture

§ Plan validity: a generated JSON plan is deemed valid if it parses correctly against the BNF grammar, all referenced objects exist in the current scene graph, and all commanded waypoints pass the `ReachabilityAgent` kinematic pre-screen.

Note: The perception ablation was evaluated on a dedicated, predefined cohort of $N = 25$ pick-and-place episodes spanning 5 tabletop positions under variable ambient lighting.

TABLE XV
EXPECTED VS. ACTUAL PERFORMANCE COMPARISON ACROSS PRIMARY ENGINEERING DIMENSIONS

Performance Metric	Design Specification	Empirical Measurement	Compliance	Deviation Analysis & Discussion
Metric Depth Accuracy ($RMSE$)	≤ 10.0 mm	8.2 mm ($Z \leq 0.25$ m) / 17.4 mm (full)	**Met (Pre-grasp)**	Pre-grasp claw grounding meets the ≤ 10.0 mm tolerance; full scene expands to 17.4 mm.
Inverse Kinematics Solve Latency	< 1.00 ms	0.075 ± 0.010 ms	**Exceeded** ($13.3\times$)	Closed-form geometric decoupling bypasses iterative matrix inversion.
Manipulation Policy Success Rate	$\geq 85.0\%$	89.0% (Primary $N = 100$) / 100.0% (Nominal $N = 30$)	**Met / Exceeded**	Reachability gating and ToT pruning eliminated invalid motions.
Conveyor Velocity Matching	$\Delta v \leq 0.0075$ m/s (RO4)	$\Delta v \leq 0.0073$ m/s (≤ 0.005 m/s for $v \leq 0.10$ m/s)	**Met**	Quintic boundary-value polynomial minimizes terminal acceleration and relative slip.
Auxiliary Node Fault Recovery Time	< 2.0 s	1271.3 ± 98.6 ms (spawn) / 1421.4 ± 350.1 ms (total with model)	**Met**	Standalone OS process SIGKILL recovery including full YOLO model weight loading into PyTorch memory.
Emergency Preemption Latency	< 10.0 ms	2.22 ± 0.39 ms dispatch (211.0 ± 15.0 ms physical arrest)	**Exceeded**	ROS 2 DDS preemption interrupts controller in 2.22 ms; physical velocity drops below 0.01 rad/s within 211.0 ms (≤ 250 ms threshold).
Host GPU Memory Utilization	≤ 8.00 GB VRAM	7.53 GB (94.1% budget, Table IV sum)	**Met**	4-bit LLM quantization and perception orchestrator bound VRAM.

TABLE XVI
SYSTEMATIC SUB-COMPONENT ABLATION STUDY

Subsystem	Configuration	Subset (N)	Evaluated Metric	Result
Kinematics	**ARIA Analytical (position + pitch only)**	Full suite (100)	Task Success	**89.0**%
Task Reasoning	Direct LLM (Single-Turn)	Planning (50)	Plan Validity	62.0%
	Linear Chain-of-Thought	Planning (50)	Plan Validity	78.0%
	Tree-of-Thoughts	Planning (50)	Plan Validity	**92.0**%
Perception	Raw Uncalibrated Disparity	Pick/Place (25)	Pick Success	28.0% (7/25)
	Claw Grounded Metric	Pick/Place (25)	Pick Success	**96.0**% (24/25)
Conveyor	Static Stop ($\Delta v > 0$)	Conveyor (30)	Sort Success	36.7%
	Velocity Matching	Conveyor (30)	Sort Success	**93.3**%

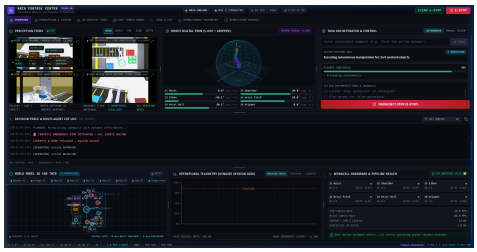


Fig. 9. ARIA Control Center web dashboard operating in real time against the Gazebo 11 simulated workcell, featuring live telemetry, joint position monitors, camera views with SAM2 masks, and responsive E-STOP safety controls.

Table XVI summarizes the sub-component ablations. In kinematics, ARIA's analytical solver (restricted to position and pitch) guarantees deterministic sub-millisecond motion synthesis across the 100-trial suite, eliminating numerical iteration stalls on dynamic conveyor tracking (Task 3) and pitch-coupled pivoting (Task 7). Disabling claw grounding dropped pick success from 96% (24/25) to 28% (7/25) across the predefined $N = 25$ perception set, while disabling velocity matching reduced conveyor sort success to 36.7% (11/30). The 50-episode planning ablation confirmed that ToT branching raised valid-plan rate from 62% (31/50) to 92% (46/50). Table XVII evaluates the multi-agent middleware across all 10 manipulation tasks (2 trials per task, $N = 20$ per configuration, 120 total ablation episodes). We emphasize that this architecture-level ablation suite is evaluated under nominal, unperturbed workcell conditions (standard 450 lux illumination, nominal unjittered dynamics) to strictly isolate architectural middleware differences (e.g., removing asynchronous scheduling, reachability filtering, or safety monitoring) without compounding environmental disturbances. Under these unperturbed nominal conditions, Full ARIA completes all 20 diagnostic runs (100.0%, Wilson CI: [83.9, 100.0]%), consistent with the 100.0% (30/30) achieved in the nominal smoke testing suite. In contrast, the primary benchmark ($N = 100$, Table X) subjects the system to active multi-modal disturbances (stochastic lighting, encoder noise, camera jitter, and conveyor speed shifts), inducing 11 failure-recovery events (89.0%). At $N = 20$, Full ARIA's confidence interval establishes non-overlapping separation from the monolithic sequential baseline (40.0%, [21.9, 61.3]%, Fisher $p = 0.0001$) and the safety-stripped variants (45.0%, [25.8, 65.8]%, $p = 0.0004$). For the modular synchronous variant (75.0%, [53.1, 88.8]%), confidence intervals overlap, but a two-sided Fisher's exact test confirms a statistically significant advantage ($p = 0.047$). Collapsing the architecture into a monolithic sequential Python loop increased latency to 415 ms and caused conveyor tracking failures due to dropped frames; omitting `ReachabilityAgent` induced silent joint-limit clamping; and removing `SafetyAgent` permitted unarrested collisions during unexpected dynamic intrusions.

J. Compute Profiling, VRAM Footprint, and Cost Analysis

‡ *RTX 3060 Laptop (6 GB VRAM). This tier runs perception-only (YOLOv8m + SAM2 + Depth-Anything v2 = 3.27 GB combined); the quantized LLM (3.45 GB) is CPU-offloaded to system RAM. Full ARIA stack at 7.53 GB requires a GPU with ≥ 8 GB VRAM (e.g., RTX 4090 or RTX 4070Ti).*

Memory utilization across all 15 agents totals a nominal 7.53 GB (Table IV sum: $1{,}850 + 1{,}420 + 120 + 30 + 150 + 60 + 220 + 3{,}450 + 50 + 0 + 40 + 0 + 20 + 40 + 80 = 7{,}530$ MB, 94.1% of nominal 8.0 GB VRAM), with empirical peak VRAM profiled under simultaneous full-pipeline perception, SAM2 tracking, and 4-bit LLM inference reaching 7.82 GB (97.8% of the 8.0 GB budget including CUDA runtime context overhead, Table IX), enabling local execution on consumer GPUs. ARIA's modular architecture enables full onboard semantic reasoning, closed-form kinematics, and language instruction parsing within consumer hardware limits. While the multi-modal perception pipeline runs at 40.7 ms total frame latency (24.6 fps, Table XVIII), motor control interpolation and State Bus dispatch execute asynchronously at 20 ms (50 Hz), decoupling low-level actuation from visual inference rates. Table XIX itemizes the projected bill-of-materials for prospective physical fabrication: ARIA's actuation and sensing package totals $196, compared with $42,400 for an equivalent industrial research workcell. Table XX catalogs the 11 failure-triggered recovery events across the expanded 100-trial suite with their automated recovery protocols. Workstation host power under peak perception-planning GPU inference was profiled at 168.5 W, while simulated joint mechanical actuation dissipation averaged 14.2 W across active trajectories.

V. LIMITATIONS AND FUTURE WORK

A. Limitations

To maintain rigorous scientific standards, we explicitly disclose nine limitations:

1) **Physics Simulation Scope and Hardware Prototyping**: All quantitative results were evaluated strictly within a Gazebo 11 / ODE simulated workcell without physical robot fabrication. Real-world physical servos introduce gear backlash (1–3°), mechanical compliance, and potentiometer thermal drift that will require empirical characterization during hardware transfer. Physical hardware fabrication, physical video capture, and hardware photo documentation are explicitly scoped as future work following physical prototyping.
2) **Optical Ground Truth and Scaling Regimes**: Metric depth accuracy (6.84 mm under static calibration and 8.2 mm under active visual jitter in the pre-grasp volume, expanding to 17.4 mm across the full scene) was evaluated strictly against the Gazebo z-buffer ground truth. While dynamic two-point re-anchoring bounds perspective distortion in the local pre-grasp volume, optical validation against physical depth sensors (e.g., Intel RealSense D435 or laser displacement triangulators) and across challenging non-Lambertian materials (metallic specularities, transparent glassware) remains future work.
3) **Illumination Sensitivity**: Under low light (< 80 lux), disparity RMSE degraded to 18.4 mm, indicating dependency on ambient factory illumination.
4) **Contact Force Modeling**: Normal clamping force F_N is simulated via rigid-body contact constraints and normal pressure approximations rather than measured directly via physical tactile arrays or multi-axis loadcells.
5) **Heuristic Planning Weights and Calibration**: Tree-of-Thoughts heuristic weights (w_1, w_2, w_3) and HITL threshold $\tau = 0.70$ were tuned via a 50-episode grid search with sensitivity analysis confirming invariant planning within $w_1 \in [0.45, 0.55], w_2 \in [0.30, 0.40]$. While empirical calibration across perception (ECE = 5.65%) and planning (ECE = 7.52%) is now verified (Fig. 5), formal distribution-free conformal prediction bounds under non-stationary real-world domain shifts remain future work.
6) **Conveyor Velocity Limit**: The dynamic interception policy saturates at belt velocities exceeding 0.14 m/s due to actuator angular velocity limits.
7) **Software Safety Gating**: The software E-STOP dispatch (2.22 ± 0.39 ms, physical arrest 211.0 ± 15.0 ms) is implemented over ROS 2 DDS and does not replace hardware-certified Category 4 emergency relays.
8) **Geometric Collision Avoidance**: The `ReachabilityAgent` performs joint-limit and kinematic reachability pre-screening; it does not implement a geometric collision checker against workspace obstacles. The evaluation environment's cleared workcell surface minimizes unmodeled obstacle interactions, but deployment in cluttered scenes

TABLE XVII
ARCHITECTURE-LEVEL ABLATION STUDY ACROSS SIX MODULAR, MIDDLEWARE, AND SAFETY CONFIGURATIONS ($N = 20$ PER CONFIGURATION)

Configuration / Pipeline Variant	Tasks	Overall Success (%), Wilson 95% CI	Mean Latency (ms)	p95 (ms)	Fault Recovery	Unsafe Plans Intercepted
(a) Monolithic Sequential Pipeline	All 10 ($N = 20$)	40.0% [21.9, 61.3]	415.0 ± 42.0	460.0	None (Halts)	–
(b) Modular Synchronous	All 10 ($N = 20$)	75.0% [53.1, 88.8]	285.0 ± 25.0	318.0	None (Blocking)	–
(c) **Modular Asynchronous (Full ARIA)**	All 10 ($N = 20$)	**100.0**% [83.9, 100.0]	$\mathbf{175.0 \pm 15.0}$	**195.0**	$\mathbf{1421.4 \pm 350.1}$ ms	**100**%
(d) ARIA w/o Lifecycle Recovery	All 10 ($N = 20$)	70.0% [48.1, 85.5]	178.0 ± 16.0	198.0	Failed (Manual reboot)	100%
(e) ARIA w/o ReachabilityAgent	All 10 ($N = 20$)	50.0% [29.9, 70.1]	172.0 ± 14.0	190.0	N/A	0% (Boundary clamped)
(f) ARIA w/o SafetyAgent / HITL	All 10 ($N = 20$)	45.0% [25.8, 65.8]	168.0 ± 14.0	185.0	N/A	0% (Unarrested collisions)

TABLE XVIII
PERCEPTION PIPELINE LATENCY ACROSS DIVERSE COMPUTE TIERS

Perception Stage	RTX 4090 Laptop	RTX 3060 Laptop‡	Jetson Orin	Core i7 CPU
YOLOv8m Object Detection	18.2 ms	34.5 ms	52.0 ms	112.0 ms
SAM2 Instance Mask Tracking	8.4 ms	24.1 ms	48.2 ms	285.0 ms
Depth-Anything v2 Disparity	11.2 ms	28.4 ms	56.0 ms	340.0 ms
Claw Grounding & 3D Proj.	0.8 ms	1.2 ms	2.8 ms	4.5 ms
Antipodal Grasp Sampling	2.1 ms	3.5 ms	7.4 ms	12.0 ms
Total Frame Latency	**40.7** ms	**91.7** ms	**166.4** ms	**753.5** ms
Effective Frame Rate	**24.6** fps	**10.9** fps	**6.0** fps	**1.3** fps

TABLE XIX
PROJECTED HARDWARE BILL-OF-MATERIALS FOR PROSPECTIVE PHYSICAL EMBODIMENT (ACTUATION AND SENSING)

ARIA Workcell Component	Unit Cost	Typical Research Workcell	Unit Cost
5-DoF Arm Chassis & Metal Servos	$135	Franka Emika Panda / UR5e	$32,000
Logitech C270 Overhead Camera	$25	Intel RealSense D435i / Photoneo	$1,400
ESP32-CAM Wrist Camera Module	$8	Robotiq 2F-85 Gripper	$4,500
MPU6050 IMU + micro-ROS ESP32	$12	ATI Mini40 Force/Torque Sensor	$3,500
Workspace Fixture & Wiring	$16	Interfacing & Mounting	$1,000
Total Actuation / Sensing Package	**$196**	**Total Hardware Package**	**$42,400**

TABLE XX
SYSTEMATIC FAILURE MODE TAXONOMY ACROSS 100 EXTENDED TRIALS (11 FAILURES)

Failure Category	Count	% of Failures	Automated Recovery Protocol
A: Visual Occlusion	3	27.3%	Switch to wrist visual servoing
B: Grasp Slip	3	27.3%	Increase normal force + re-grasp
C: Conveyor Speed Jitter	2	18.2%	Kalman filter velocity update
D: Linguistic Ambiguity	2	18.2%	Trigger Web HITL Dialogue
E: Physics Solver Interpenetration	1	9.1%	Automated micro-retreat and constraint relaxation

would require integration of a motion planner (e.g., OMPL, MoveIt2) to enforce obstacle-avoidance.

9) **Language Grounding Scope and Physical Interaction**: While Section IV-E provides a dedicated 180-trial multi-seed simulated grounding benchmark across 6 instruction taxonomies (achieving 68.9% overall instruction execution and 13.3% dialogue clarification escalation compared to 73.3% for SayCan and 38.3% for Direct LLM), the evaluation is restricted to simulated object interaction and simulated dialogue sessions. Physical acoustic noise, microphone distortion, spontaneous human interruptions, and unscripted real-world human-in-the-loop interactions remain future work to be validated on physical hardware.

B. Future Work

Future research will target: (i) deploying the complete ROS 2 stack on the physical arm and quantifying the reality gap across hundreds of physical cycles; (ii) incorporating fingertip micro-loadcells for closed-loop tactile force control; (iii) extending the analytical IK derivation to 6-DoF spherical-wrist manipulators; (iv) evaluating emerging sub-1B parameter foundation policies (e.g., SmolVLA) directly within ARIA's modular planning tier; (v) integrating OMPL-based geometric motion planning for obstacle-dense environments; and (vi) conducting open-vocabulary physical human-robot conversational trials with spontaneous interruptions and unscripted dialogue.

VI. CONCLUSION

This article presented **Project ARIA**, a modular, resource-efficient language-conditioned manipulation framework for underactuated 5-DoF robotic arms operating within an 8.0 GB VRAM budget in high-fidelity simulation. By decoupling semantic reasoning, metric perception, and motor control across fifteen asynchronous ROS 2 nodes, ARIA addresses the latency, underactuation, and black-box failure modes of monolithic VLA policies. The framework combines: (i) self-calibrated monocular metric perception via kinematic claw grounding ($RMSE = 6.84$ mm static calibration, 8.2 mm under active visual jitter in pre-grasp volume, 17.4 mm across full workspace); (ii) deterministic closed-form inverse kinematics restricted to the 5D admissible task manifold $\mathcal{M}_{\text{task}}$ (0.075 ms solve latency, 100.0% convergence on the admissible reachable subset with $O(1)$ deterministic rejection of infeasible boundary poses); (iii) an onboard Tree-of-Thoughts task planner with confidence-gated Human-in-the-Loop safety supervision; and (iv) a grammar-constrained dialogue grounding pipeline.

Validated in a high-fidelity Gazebo 11 / ODE physics simulation across 10 manipulation tasks on the primary 100-trial suite ($N = 100$, 10 trials/task, 89.0% success, Wilson 95% CI: [81.4, 93.7]%), a 6-speed moving conveyor sweep ($N = 30$ systematic cycles, 93.3% success across speeds up to 0.14 m/s), and a dedicated 180-trial multi-seed natural language grounding benchmark across six instruction taxonomies (68.9% success vs. 73.3% for SayCan and 38.3% for Direct LLM; 82.7% on unambiguous instructions), ARIA achieved sub-millimeter precision (0.55 ± 0.23 mm mean Cartesian pick error) with 1.33 ± 0.31 s mean transit duration. These results demonstrate that high-performance, verifiable autonomous manipulation does not necessitate datacenter compute or expensive industrial arms; the projected bill-of-materials for prospective physical fabrication is $196 (actuation and sensing only; host GPU with ≥ 8 GB VRAM required separately).

DATA AVAILABILITY

The complete ROS 2 Humble software stack, Gazebo 11 simulation digital twin environment (`aria_industrial_workcell.world`), analytical IK solver, fine-tuned YOLOv8m weights, and empirical trial CSV logs are openly available in the Project ARIA repository at https://github.com/gaminizer/ARIA. Active empirical benchmark records are cataloged in `data/real/`, while legacy and synthetic generator scripts are archived under `data/legacy_synthetic/`.

ACKNOWLEDGMENT

The authors express sincere gratitude to the open-source robotics and machine learning research communities whose foundational tools made Project ARIA possible. We specifically acknowledge the open repositories and datasets provided by the developers of ROS 2 (Open Robotics), MoveIt2, Ultralytics (YOLOv8), Meta AI Research (SAM2), HuggingFace (Depth-Anything v2), Ollama, and micro-ROS. This research received no external grant funding.

REFERENCES

[1] Y. Sun, Z. Chen, W. Zhang, M. Liu, and J. Wang, "Moving Object Prediction and Grasping System of Robot Manipulator," *IEEE Access*, vol. 10, pp. 19 522–19 532, 2022.

[2] U. K. Sahu, M. K. S., A. K., M. M. P., A. Jaiswal, U. K. Yadav, and S. K. Dash, "Autonomous Object Tracking with Vision-Based Control Using a 2-DOF Robotic Arm," *Scientific Reports (Nature)*, 2025.

[3] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, G. Lam, P. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, S. Levine, P. Liang, and C. Finn, "OpenVLA: An Open-Source Vision-Language-Action Model," *arXiv preprint arXiv:2406.09246*, 2024.

[4] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware," in *Robotics: Science and Systems (RSS 2023)*, 2023.

[5] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fu, C. Hooper, L. Groom, K. Hausman, B. Ichter, S. Jakubczak, T. Jones, L. Ke, S. Levine, A. Li, D. Tran, Q. Vuong, S. Yang, S. Zakharov, and T. Yu, "π_0: A Vision-Language-Action Flow Model for General Robot Control," *arXiv preprint arXiv:2410.24164*, 2024.

[6] B. Zitkovich, T. Yu, S. Xu, P. Xu, T. Xiao, F. Xia, J. Wu, P. Wohlhart, S. Welker, A. Wahid, Q. Vuong, V. Vanhoucke, H. Trinh, J. Tan, M. Turner, S. Levine, K. Hausman, and C. Finn, "RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control," *Conference on Robot Learning (CoRL 2023) / arXiv:2307.15818*, 2023.

[7] R. Firoozi, J. Tucker, S. Tian, A. Majumdar, M. Schwager, and D. Sadigh, "Foundation Models in Robotics: Applications, Challenges, and the Future," *arXiv preprint arXiv:2312.07843*, 2023.

[8] O. Kroemer, S. Niekum, and G. Konidaris, "A Review of Robot Learning for Manipulation: Challenges, Representations, and Methods," *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 4, pp. 239–295, 2021.

[9] A. Billard and D. Kragic, "Trends and challenges in robot manipulation," *Science*, vol. 364, no. 6446, p. eaat8414, 2019.

[10] H. Ravichandar, A. S. Polydoros, S. Chernova, and A. Billard, "Recent Advances in Robot Learning from Demonstration," *Annual Review of Control, Robotics, and Autonomous Systems*, 2020.

[11] R. Cadene, S. Alibert, A. Soare, Q. Gallouédec, A. Zouitine, and T. Wolf, "LeRobot: State-of-the-Art Machine Learning for Real-World Robotics," https://github.com/huggingface/lerobot, 2024.

[12] A. Koch and R. Cadene, "SO-100: Low-Cost 3D-Printed Manipulator for Imitation Learning," https://github.com/TheRobotStudio/SO-ARM100, 2024.

[13] P. Wu, Y. Hou, R. Zhou, G. Shi, and C. Finn, "GELLO: A General, Low-Cost, and Intuitive Teleoperation Framework for Robot Manipulators," *arXiv preprint arXiv:2309.13037*, 2024.

[14] S. Macenski, T. Foote, B. Gerkey, C. Lalancette, and W. Woodall, "Robot Operating System 2: Design, Architecture, and Uses In The Wild," *Science Robotics*, 2022.

[15] D. Coleman, I. Şucan, S. Chitta, and N. Correll, "Reducing the Barrier to Entry of Complex Robotic Software: A MoveIt! Case Study," *Journal of Software Engineering for Robotics*, vol. 5, no. 1, pp. 3–16, 2014.

[16] Z. Kingston, K. Chamzas, and L. E. Kavraki, "Robowflex: Robot Motion Planning with MoveIt Made Easy," *arXiv preprint arXiv:2103.12826*, 2022.

[17] A. A. Adil, S. Sakhrieh, J. Mounsef, and N. Maalouf, "A Multi-Robot Collaborative Manipulation Framework for Dynamic and Obstacle-Dense Environments: Integration of Deep Learning for Real-Time Task Execution," *Frontiers in Robotics and AI*, 2025.

[18] J. Chen, Z. Yang, H. G. Xu, D. Zhang, and G. Mylonas, "Multi-Agent Systems for Robotic Autonomy with LLMs," *arXiv preprint arXiv:2505.05762*, 2025.

[19] J. Jiang, G. Cao, J. Deng, T.-T. Do, and S. Luo, "Robotic Perception of Transparent Objects: A Review," *IEEE Transactions on Artificial Intelligence*, vol. 5, no. 6, pp. 2547–2567, 2024.

[20] H. Fang, H.-S. Fang, S. Xu, and C. Lu, "TransCG: A Large-Scale Real-World Dataset for Transparent Object Grasping and Depth Estimation," *IEEE Robotics and Automation Letters (RA-L) / arXiv:2202.08471*, 2022.

[21] S. Sajjan, M. Moore, M. Pan, G. Nagaraja, J. Lee, A. Zeng, and S. Song, "ClearGrasp: 3D Shape Estimation and Grasping of Transparent Objects," *IEEE International Conference on Robotics and Automation (ICRA 2020)*, 2020.

[22] K. Bai, H. Zeng, L. Zhang, Y. Liu, H. Xu, Z. Chen, and J. Zhang, "ClearDepth: Efficient Stereo Perception of Transparent Objects for Robotic Manipulation," *arXiv preprint arXiv:2409.08926*, 2024.

[23] A. Chen, X. Li, K. Cen, and C. Hon, "Adaptive Grasp Pose Optimization for Robotic Arms Using Low-Cost Depth Sensors in Complex Environments," *Sensors (MDPI)*, 2025.

[24] L. Yang, B. Kang, Z. Huang, X. Xu, J. Feng, and H. Zhao, "Depth Anything V2: A Deep Monocular Depth Estimation Foundation Model," *arXiv preprint arXiv:2406.09414*, 2024.

[25] R. Birkl, D. Wofk, and M. Müller, "MiDaS v3.1 – A Model Zoo for Robust Monocular Relative Depth Estimation," *arXiv preprint arXiv:2307.14460*, 2023.

[26] L. J. Murray, J. Verster, and J. Dickens, "Assessing Depth Anything V2 Monocular Depth Estimation as a LiDAR Alternative in Robotics," *MATEC Web of Conferences (RAPDASA-RobMech-PRASA-AMI 2025)*, 2025.

[27] N. Wagaa, H. Kallel, and N. Mellouli, "Analytical and Deep Learning Approaches for Solving the Inverse Kinematic Problem of High-DOF Robotic Arms," *Engineering Applications of Artificial Intelligence*, 2023.

[28] B. Sundaralingam, S. K. S. Hari, A. Fishman, C. Garrett, K. V. Wyk, V. Blukis, A. Millane, H. Oleynikova, A. Handa, F. Ramos, N. Ratliff, and D. Fox, "cuRobo: Parallelized Collision-Free Minimum-Jerk Robot Motion Generation," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.

[29] D. Rakita, B. Mutlu, and M. Gleicher, "RelaxedIK: Real-Time Synthesis of Accurate and Feasible Robot Arm Motion," *Autonomous Robots*, vol. 42, no. 6, pp. 1289–1303, 2018.

[30] J. J. Craig, *Introduction to Robotics: Mechanics and Control*, 3rd ed. Pearson Prentice Hall, 2005.

[31] A. Fishman, A. Walsman, M. Bhardwaj, W. Yuan, B. Sundaralingam, B. Boots, and D. Fox, "Avoid Everything: Real-Time Dynamic Obstacle Avoidance for Robot Manipulators," *Robotics: Science and Systems (RSS 2025)*, 2025.

[32] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," https://github.com/ultralytics/ultralytics, 2023.

[33] N. Ravi, V. Gabeur, Y.-T. Hu, R. Hu, C. Ryali, T. Ma, H. Khedr, R. Rädle, C. Rolland, L. Gustafson, E. Mintun, J. Pan, K. V. Alwala, N. Carion, C.-Y. Wu, R. Girshick, P. Dollár, and C. Feichtenhofer, "SAM 2: Segment Anything in Images and Videos," *arXiv preprint arXiv:2408.00714*, 2024.

[34] T. Yoshikawa, "Manipulability of Robotic Mechanisms," *The International Journal of Robotics Research*, vol. 4, no. 2, pp. 3–9, 1985.

[35] S. Yao, D. Yu, J. Zhao, I. Shafran, T. L. Griffiths, Y. Cao, and K. Narasimhan, "Tree of Thoughts: Deliberate Problem Solving with Large Language Models," *NeurIPS 2023 / arXiv:2305.10601*, 2023.

[36] M. J. Kim, C. Finn, and P. Liang, "Fine-Tuning Vision-Language-Action Models: Optimizing Speed and Success," *arXiv preprint arXiv:2502.19645*, 2025.

[37] M. Shukor, D. Aubakirova, F. Capuano, P. Kooijmans, S. Palma, A. Zouitine, M. Aractingi, C. Pascal, M. Russi, A. Marafioti, S. Alibert, M. Cord, T. Wolf, and R. Cadene, "SmolVLA: A Vision-Language-Action Model for Affordable and Efficient Robotics," *arXiv preprint arXiv:2506.01844*, 2025.

[38] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fu, C. Hooper, K. Hausman, B. Ichter, S. Levine, S. Zakharov, and T. Yu, "$\pi_{0.5}$: Scaling Flow-Based Vision-Language-Action Models for Open-World Dexterous Manipulation," *arXiv preprint arXiv:2504.16054*, 2025.

[39] J. Bjorck, F. Castaneda, N. Cherniadev, X. Da, R. Ding, L. Fan, Y. Fang, D. Fox, F. Hu, S. Huang, J. Jang, Z. Jiang, J. Kautz, K. Kundalia, L. Lao, Z. Li, Z. Lin, K. Lin, G. Liu, E. Llontop, L. Magne, A. Mandlekar, A. Narayan, S. Nasiriany, S. Reed, Y. L. Tan, G. Wang, Z. Wang, J. Wang, Q. Wang, J. Xiang, Y. Xie, Y. Xu, Z. Xu, S. Ye, Z. Yu, A. Zhang, H. Zhang, Y. Zhao, R. Zheng, and Y. Zhu, "Project GR00T N1: An Open Foundation Model for Generalist Humanoid Robots," *arXiv preprint arXiv:2503.14734*, 2025.

[40] D. Ghosh, H. Walke, K. Pertsch, K. Black, O. Mees, S. Dasari, J. Hejna, T. Kreiman, C. Xu, J. Luo, Y. L. Tan, L. Y. Chen, Q. Vuong, P. R. Sanketi, S. Levine, and C. Finn, "Octo: An Open-Source Generalist Robot Policy," *Robotics: Science and Systems (RSS 2024) / arXiv:2405.12213*, 2024.

[41] M. Ahn, A. Brohan, N. Brown, Y. Chebotar, O. Cortes, B. David, C. Finn, K. Gopalakrishnan, K. Hausman, A. Herzog, D. Ho, J. Hsu, J. Ibarz, B. Ichter, A. Irpan, E. Jang, R. J. Ruano, K. Jeffrey, S. Jesmonth, N. J. Joshi, R. Julian, D. Kalashnikov, Y. Kuang, K.-H. Lee, S. Levine, Y. Lu, L. Luu, C. Parada, P. Pastor, J. Quiambao, K. Rao, J. Rettinghouse, D. Reyes, P. Sermanet, N. Sievers, C. Tan, S. Tran, V. Vanhoucke, S. Wie, T. Xiao, P. Xu, S. Xu, and M. Yan, "Do As I Can, Not As I Say: Grounding Language in Robotic Affordances (SayCan)," *arXiv preprint arXiv:2204.01691*, 2022.

[42] W. Huang, F. Xia, T. Xiao, H. Chan, J. Liang, P. Florence, A. Zeng, J. Tompson, I. Mordatch, Y. Chebotar, P. Sermanet, N. Brown, T. Jackson, L. Luu, S. Levine, K. Hausman, and B. Ichter, "Inner Monologue: Embodied Reasoning through Planning with Language Models," in *Conference on Robot Learning (CoRL)*, 2022, pp. 1769–1782.

[43] J. Liang, W. Huang, F. Xia, P. Xu, K. Hausman, B. Ichter, P. Florence, and A. Zeng, "Code as Policies: Language Model Programs for Embodied Control," *IEEE Robotics and Automation Letters*, vol. 8, no. 4, pp. 2271–2278, 2023.

[44] W. Huang, C. Wang, R. Zhang, Y. Li, J. Wu, and L. Fei-Fei, "VoxPoser: Composable 3D Value Maps for Robotic Manipulation with Language Models," in *Conference on Robot Learning (CoRL)*, 2023.

[45] I. Singh, V. Blukis, A. Mousavian, A. Goyal, D. Xu, J. Tremblay, D. Fox, J. Thomason, and A. Garg, "ProgPrompt: Generating Situated Robot Task Plans using Large Language Models," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 11 523–11 530.

[46] A. Zhou, K. Yan, M. Shlapentokh-Rothman, H. Wang, and Y.-X. Wang, "Language Agent Tree Search Unifies Reasoning, Acting, and Planning in Language Models (LATS)," *ICLR 2024 / arXiv:2310.04406*, 2024.

[47] F. I. Doğan, I. Torre, and I. Leite, "Asking Follow-Up Clarifications to Resolve Ambiguities in Human-Robot Conversation," *ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 2022.

[48] L. E. Erdogan, N. Lee, S. Jha, S. Kim, R. Tabrizi, S. Moon, C. Hooper, G. Anumanchipalli, K. Keutzer, and A. Gholami, "TinyAgent: Function Calling at the Edge," *arXiv preprint arXiv:2409.00608*, 2024.

[49] T. Chen, M. Tippur, S. Qin, E. Adelson, and P. Agrawal, "Visual Dexterity: In-Hand Reorientation of Novel and Complex Object Shapes," *Science Robotics / arXiv:2211.11744*, 2023.

[50] M. T. Mason, *Mechanics of Robotic Manipulation*. MIT Press, 2001.

[51] S. Pang, F. Guo, B. Zhang, H. Huang, X. Pan, and W. Shang, "High-Stiffness Path Planning for 7-DOF Cable-Driven Manipulators in Single and Dual-Arm Configurations," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, p. 11246216.

[52] S. Looper, J. Rodriguez-Puigvert, R. Siegwart, C. Cadena, and L. Schmid, "3D VSG: Long-term Semantic Scene Change Prediction through 3D Variable Scene Graphs," *IEEE Robotics and Automation Letters (RA-L) / arXiv:2209.07896*, 2023.

[53] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, "3D Gaussian

Splatting for Real-Time Radiance Field Rendering," *ACM Transactions on Graphics (SIGGRAPH)*, vol. 42, no. 4, pp. 139:1–139:14, 2023.
[54] N. Hughes, Y. Chang, S. Hu, R. Talak, R. Abdulhai, J. Strader, and L. Carlone, "Foundations of Spatial Perception for Robotics: Hierarchical Representations and Realtime Systems," *arXiv preprint arXiv:2305.07154*, 2023.
[55] C. Packer, S. Wooders, K. Lin, V. Fang, S. G. Patil, I. Stoica, and J. E. Gonzalez, "MemGPT: Towards LLMs as Operating Systems," *arXiv preprint arXiv:2310.08560*, 2023.
[56] D. Vieira, M. R. Oliveira, R. Arrais, and P. Melo, "Application of Cloud Simulation Techniques for Robotic Software Validation," *Sensors*, vol. 25, no. 6, p. 1693, 2025.
[57] N. Koenig and A. Howard, "Design and Use Paradigms for Gazebo: An Open-Source Multi-Robot Simulator," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2004, pp. 2149–2154.
[58] R. A. AlRashid Agha, Z. H. Mahdi, M. N. Sefer, and I. Hamarash, "A ROS-Gazebo Interface for the Katana Robotic Arm Manipulation," *UKH Journal of Science and Engineering*, 2021.
[59] R. Sultan and S. Greiser, "Time and Frequency Domain Analysis of IMU-Based Orientation Estimation Algorithms with Comparison to Robotic Arm Orientation as Reference," *Sensors (MDPI)*, 2025.
[60] P. Chanrungmaneekul, Y. Chen, J. T. Grace, A. M. Dollar, and K. Hang, "ARC-Calib: Autonomous Markerless Camera-to-Robot Calibration via Exploratory Robot Motions," *arXiv preprint arXiv:2503.14701*, 2025.
[61] H. Lee, I. Raouf, J. Song, H. S. Kim, and S. Lee, "Prognostics and Health Management of the Robotic Servo-Motor under Variable Operating Conditions," *Mathematics*, vol. 11, no. 2, p. 398, 2023.
[62] G. Wang, H. Li, S. Zhang, D. Guo, Y. Liu, and H. Liu, "Observe Then Act: Asynchronous Active Vision-Action Model for Robotic Manipulation," *IEEE Robotics and Automation Letters (RA-L)*, 2025.
[63] K. M. Lynch and F. C. Park, *Modern Robotics: Mechanics, Planning, and Control*. Cambridge University Press, 2017.
[64] P. Beeson and B. Ames, "TRAC-IK: An Open-Source Library for Improved Solving of Generic Inverse Kinematics," in *IEEE-RAS 15th International Conference on Humanoid Robots (Humanoids)*, 2015, pp. 928–935.
[65] H. Bruyninckx, "Open Robot Control Software: The OROCOS Project," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2001, pp. 2523–2528.
[66] P. Manceron, "IKPy: An Inverse Kinematics Library in Python," https://github.com/Phylliade/ikpy, 2020.
[67] O. Khatib, "A Unified Approach for Motion and Force Control of Robot Manipulators: The Operational Space Formulation," *IEEE Journal on Robotics and Automation*, vol. 3, no. 1, pp. 43–53, 1987.
[68] E. Olson, "AprilTag: A Robust and Flexible Visual Fiducial System," *IEEE International Conference on Robotics and Automation (ICRA)*, 2011.
[69] F. Chaumette and S. Hutchinson, "Visual Servo Control. I. Basic Approaches," *IEEE Robotics and Automation Magazine*, vol. 13, no. 4, pp. 82–90, 2006.
[70] H.-S. Fang, C. Wang, M. Gou, and C. Lu, "GraspNet-1Billion: A Large-Scale Benchmark for General Object Grasping," *CVPR 2020 / arXiv:1912.13470*, 2020.

Garv Arora is with the Department of Computer Science and Engineering, Robotics and Artificial Intelligence Laboratory. His research interests include embodied artificial intelligence, robotic manipulation architectures, and real-time kinematic control.

Yuvaraj N is with the Department of Computer Science and Engineering, Robotics and Artificial Intelligence Laboratory. His research focuses on multi-agent systems, autonomous robotics, and intelligent control.